



CLOSING THE THREE-GAP FRAMEWORK: A SYSTEMATIC REVIEW OF EVALUATION, GOVERNANCE, AND IMPLEMENTATION CHALLENGES IN CLINICAL AI DEPLOYMENT *

Master's Thesis

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ABSTRACT

Despite rapid growth in research on artificial intelligence in healthcare, the literature remains fragmented across disciplinary silos that each address isolated facets of how AI is evaluated and governed, with little synthesis of how these dimensions interact in real-world deployment. This review identifies a three-gap framework as its central conceptual contribution, highlighting three interlocking structural deficits that constrain the safe and equitable translation of clinical AI from pilot projects to routine care: the **ecological validity gap** between controlled validation studies and the realities of clinical practice; the **governance capacity gap** between the institutional preconditions emerging regulatory frameworks assume and the organizational readiness health systems possess; and the **pilot trap**—the systematic failure of clinically-validated AI systems to transition from short-term demonstration projects into sustained components of routine care, often mediated by post-market performance drift and absent accountability infrastructure for continuously-learning systems. A systematic search was conducted across PubMed, Semantic Scholar, arXiv, and DBLP for peer-reviewed publications and preprints published between 2020 and 2026, yielding a final corpus of 311 papers that were synthesized thematically. The synthesis reveals that these three gaps are not independent. Rather, they reinforce one another in a self-perpetuating cycle: ecologically weak validation enables deployment into institutions lacking sufficient governance capacity, which in turn are unable to sustain the post-deployment monitoring and accountability structures that the pilot trap demands. Current evaluation practices remain dominated by retrospective benchmarking on curated datasets; participation of patients, clinicians, and communities in AI governance remains largely episodic rather than institutionalized; and the most promising regulatory initiatives are those that treat evaluation, governance, and implementation as components of a continuous lifecycle rather than as separate phases. A growing body of literature on ecologically grounded evaluation protocols, governance maturity models, and post-deployment monitoring frameworks suggests that this cycle can be broken, but only through integrated approaches that connect technical evaluation, institutional governance, and implementation practice. The review argues that safe and equitable health AI deployment requires moving beyond pre-market authorization alone. It calls for (i) continuous post-market performance monitoring using standardized real-world metrics; (ii) institutional capacity-building as a prerequisite for sustainable AI integration; and (iii) institutionalized participatory mechanisms that ensure AI systems reflect the needs and values of the communities they serve. The review concludes by outlining the most pressing unanswered questions at the evaluation–governance–implementation interface and proposes a research agenda for closing the three identified gaps.

Keywords: health AI, AI evaluation, algorithmic governance, clinical AI deployment, adaptive regulation, participatory governance, implementation science, post-market surveillance, ecological validity

1 Introduction

1.1 Background and Motivation

Health artificial intelligence has transitioned from a research curiosity to a clinical reality. As of 2024, the United States Food and Drug Administration (FDA) had authorized nearly 1,000 AI-enabled medical devices spanning

diagnostic radiology, cardiac monitoring, ophthalmology, and oncology applications [1]. The United Kingdom's National Health Service (NHS) has similarly deployed AI tools across multiple trusts, including AI-assisted diabetic retinopathy screening programs that have screened over one million patients and algorithmic triage systems embedded within primary care pathways [nhs2024ai, price2024nhs]. In the United States, Epic Systems — one of the largest electronic health record vendors — has integrated large language models into its ambient documentation and clinical inbox summarization platforms, deploying these tools across hundreds of hospital systems and raising novel questions about institutional liability, data governance, and continuous performance monitoring [epic2024ilm, 2]. The scale of deployment is accelerating: AI-powered digital health applications attracted over USD 20 billion in venture capital funding in 2023 alone, and major hospital networks in North America, Europe, and East Asia have established dedicated AI governance committees within the past three years.

Despite this rapid uptake, a persistent pattern has emerged in the literature. Clinical AI applications frequently succeed in pilot studies but fail to become sustained components of routine clinical care. In practice, this pattern takes a recognisable shape: an AI system may demonstrate strong technical performance and promising clinical outcomes during a short-term trial, only to be quietly abandoned, partially scaled, or never integrated into routine workflows once the pilot concludes — a phenomenon that Tian et al. [3] term the **pilot trap**. This gap between demonstrated efficacy in controlled settings and real-world effectiveness is not primarily a technical problem. As the reviewed literature reveals, it is deeply rooted in governance deficits: unclear accountability structures, absent post-market monitoring mechanisms, insufficient organizational capacity, and regulatory frameworks designed for static medical devices that cannot accommodate continuously evolving AI systems [4, 5]. The consequence is a growing corpus of deployed AI systems whose long-term clinical impact remains unevaluated, whose failure modes are poorly documented, and whose governance rests on assumptions of institutional readiness that have not yet been met in most health systems.

1.2 Scientific Context and Knowledge Gap

A growing body of literature has examined individual facets of health AI deployment. Several reviews have addressed **AI evaluation methodologies** in isolation, focusing on metric design, benchmarking datasets, and technical validation protocols [see e.g. 5]. Others have examined **regulatory approaches** to AI, including the FDA's Software Pre-Certification (Pre-Cert) Program and the European Union's AI Act [1]. Still others have explored **ethical and participatory dimensions** of clinical AI, emphasizing the need for community engagement, transparency, and accountability [6]. Recently, Hussein et al. [7] proposed a maturity model for AI governance in healthcare, offering a structured assessment framework that maps organizational readiness across multiple governance dimensions. Similarly, palama2026auditing presented a four-layer auditing and monitoring framework spanning bias detection, explainability, performance surveillance, and regulatory compliance, linking continuous monitoring to institutional decision-making hierarchies.

However, no single review has yet synthesized these three streams — **evaluation methods, governance structures, and implementation outcomes** — through a unified analytical lens that takes seriously the specific challenges of real-world clinical deployment. Existing reviews tend to treat evaluation as a purely technical exercise and governance as a purely regulatory exercise, failing to account for the organizational, social, and temporal dynamics that determine whether an AI system ultimately benefits patients. Tian et al. [3] have described this disconnect explicitly, demonstrating through an 18-month longitudinal study in China that the failure of AI systems to progress from pilot to routine care is attributable not to algorithmic inadequacy but to institutional bottlenecks: fragmented accountability, underfunded post-deployment infrastructure, and governance mechanisms that are misaligned with the iterative nature of adaptive AI systems. This review addresses that gap by systematically interrogating the intersection of evaluation practice, governance architecture, and implementation evidence in health AI.

1.3 Research Question and Objectives

The central research question guiding this review is:

What methods and frameworks exist for evaluating health AI systems in real-world deployment contexts, and how do governance structures shape their implementation, sustainability, and societal impact?

This question is operationalized through three sub-questions:

- (I) **Evaluation methods:** Which evaluation frameworks, metrics, and validation methodologies are used to assess the performance, reliability, robustness, and safety of clinical AI systems in post-deployment settings? How do these methods address bias detection, algorithmic drift, and generalizability across diverse patient populations?

- (II) **Governance structures:** What regulatory, organizational, and participatory governance mechanisms have been proposed or implemented to ensure accountability, transparency, and equity in clinical AI deployment? How do these mechanisms interact across the regulatory lifecycle (pre-market authorization, post-market surveillance, adaptive updates)?
- (III) **Implementation outcomes and barriers:** What are the documented barriers to sustainable clinical AI implementation, and what organizational, technical, and social factors determine whether AI systems transition from pilot projects to routine care? What does the evidence say about real-world clinical impact?

1.4 Scope and Delimitation

This review focuses on peer-reviewed empirical and conceptual literature published between 2020 and 2026, covering AI systems deployed in or evaluated for direct or indirect clinical care — including diagnostic AI, predictive modeling, clinical decision support, digital phenotyping, and AI-assisted therapeutic applications. Studies limited to pre-clinical or animal model settings, or focusing exclusively on administrative optimization without direct clinical relevance, are excluded. The review further delimits its scope to human health AI applications, thereby excluding veterinary, environmental, or non-clinical AI deployments, as these latter domains involve distinct regulatory regimes, evaluation criteria, and societal implication profiles that fall outside the scope of the present analysis.

The review draws on a curated corpus extracted from PubMed, Semantic Scholar, arXiv, and Google Scholar using a keyword set organized across five thematic axes: *AI Evaluation and Validation*, *Participatory Governance*, *Adaptive Regulation*, *Evidence and Implementation*, and *Clinical Health AI*. A full description of the search strategy and selection criteria is provided in Section 2.

1.5 Report Structure

The remainder of this report is organized as follows. Section 2 describes the systematic search strategy, selection criteria, and analytical approach. Section 3 presents the synthesized findings across the thematic dimensions defined above, followed by a thematic application section providing deep analysis of four cornerstone papers, an empirical content analysis quantifying the three-gap framework, and a regulatory competitive analysis comparing major governance frameworks. Section 8 discusses the implications of these findings for research, policy, and clinical practice, and articulates the key limitations of this review. Section 9 offers a concluding synthesis and a research agenda. Supplementary materials, including the full PRISMA flow diagram and the annotated bibliography, are provided in Appendix B.

2 Methods

This section has two components. First, it describes the automated literature collection pipeline built to support this review — the technical and conceptual architecture that enabled systematic coverage of a rapidly evolving literature. Second, it describes the systematic review methodology itself: the search strategy, selection criteria, quality assessment, and analytical framework applied to the gathered corpus.

2.1 Pipeline Design and Implementation

The search was conducted via a semi-automated AI-assisted pipeline querying four databases (arXiv, PubMed, Semantic Scholar, DBLP) daily across five thematic axes spanning AI evaluation, participatory governance, adaptive regulation, evidence and implementation, and clinical health AI. Papers were scored using a multi-dimensional composite scoring algorithm and prioritized for human review. Full technical specifications — including the pipeline architecture, scoring dimensions, deployment scripts, and Telegram integration — are provided in Appendix B.4.

2.2 Search Strategy and Information Sources

For the systematic review itself, papers were retrieved from the pipeline database covering the period January 2020 to April 2026. The search was replicated across four electronic databases to ensure comprehensive coverage: **PubMed** (MEDLINE), **Semantic Scholar**, **arXiv**, and **Google Scholar**. The search query was constructed around the five thematic axes described above, each comprising controlled vocabulary terms and free-text keywords:

- (I) **AI Evaluation and Validation** — “AI evaluation,” “AI validation,” “algorithmic auditing,” “model evaluation,” “bias detection,” “robustness testing,” “post-deployment monitoring,” “continuous evaluation,” “explainability,” “safety evaluation.”

- (II) **Participatory Governance** — “participatory governance,” “stakeholder engagement,” “citizen participation,” “algorithmic accountability,” “democratic AI,” “community-centered AI.”
- (III) **Adaptive Regulation** — “adaptive governance,” “agile regulation,” “responsive regulation,” “regulatory sandboxes,” “living labs,” “evidence-based regulation,” “continuous compliance.”
- (IV) **Evidence and Implementation** — “evidence-based policy,” “implementation science,” “knowledge translation,” “policy uptake,” “evidence gaps.”
- (V) **Clinical Health AI** — “clinical AI,” “medical AI,” “clinical decision support,” “diagnostic AI,” “predictive modeling in healthcare,” “digital health,” “AI safety in healthcare.”

Search strings were adapted for each database using Boolean operators (AND/OR) and database-specific syntax. No language restrictions were applied at the search stage; however, only English-language publications were retained after abstract screening. Preprints and peer-reviewed articles were both included given the rapid evolution of the field.

2.3 Selection Criteria

Studies were included if they met all of the following criteria:

- **Population:** Human health AI systems deployed in clinical, sub-clinical, or community health settings; or AI systems evaluated for direct or indirect clinical application.
- **Intervention:** Any AI-based intervention — including diagnostic AI, predictive modeling, clinical decision support systems, natural language processing for clinical text, wearable sensor analytics, and AI-assisted therapeutic applications.
- **Outcomes:** Studies reporting on evaluation metrics, governance structures, implementation barriers, regulatory frameworks, ethical considerations, or societal impact of health AI.
- **Study design:** Peer-reviewed empirical studies (RCTs, cohort studies, case studies, cross-sectional studies), systematic reviews, scoping reviews, conceptual papers, and policy analyses. Animal studies, purely computational studies without clinical application, and publications in predatory journals were excluded.
- **Date range:** Published between 2020 and 2026.

2.4 Quality Assessment

Given the heterogeneous study designs included in this review, a modular quality appraisal approach was adopted. For empirical clinical studies, the **JBI Critical Appraisal Checklist** appropriate to the study design was applied. For conceptual and policy papers, a bespoke quality rubric assessed: (1) explicit statement of the research question or analytical aim; (2) transparency in methods (search strategy, selection criteria, data sources); (3) balanced representation of evidence including acknowledgment of limitations; and (4) clarity and logical coherence of argumentation. Studies were not excluded on the basis of quality alone; rather, quality appraisal scores weighted the interpretive confidence assigned to individual findings during synthesis.

2.5 Data Extraction and Analysis

Data were extracted using a standardized extraction form capturing: study metadata (authors, year, journal, country); study design; AI application domain; evaluation methods and metrics; governance and regulatory mechanisms described; implementation outcomes reported; and thematic codes assigned. Extraction was performed by the primary reviewer and verified by a secondary reviewer; discrepancies were resolved by discussion.

Analysis followed a **thematic synthesis** approach, combining inductive coding of primary findings with a deductive framework organized around the three analytical dimensions defined in the research question: (i) evaluation methods and metrics; (ii) governance and regulatory frameworks; and (iii) implementation outcomes and barriers. Results are reported narratively organized by theme, with summary tables providing an overview of the included studies and their key characteristics.

A PRISMA flow diagram is provided in Appendix B.1. The review protocol was designed and reported following a modified PRISMA framework adapted for thematic synthesis of heterogeneous evidence.

3 Results

From an initial corpus of approximately 2,700 publications identified through systematic search across arXiv, PubMed, Semantic Scholar, and DBLP, 311 papers were retained after deduplication, abstract screening, and full-text assessment against inclusion and exclusion criteria. Of these, 78 (25.1%) were designated Priority 3 (composite score ≥ 7.0) and form the primary evidence base; the remainder were retained as supporting literature. The reviewed literature reveals three overarching findings at the intersection of health AI evaluation and governance: a pervasive **ecological validity gap** in evaluation methodology; a structural **governance capacity deficit** in health systems; and a systematic **pilot-to-routine transition failure** rooted in organizational rather than technical barriers. These three findings are analytically interdependent rather than independent — the ecological validity gap generates misleading validation evidence that reduces the urgency of governance capacity-building; governance frameworks built on that evidence are structurally misaligned with real-world complexity; and that misalignment produces the pilot trap as a downstream consequence.

3.1 Evaluation Methods: The Ecological Validity Gap

3.1.1 Limited External Validation and Retrospective Bias

A consistent methodological limitation across the reviewed corpus is the predominance of single-site, retrospective evaluation designs. Studies evaluating deep learning models for medical imaging — including models for colorectal cancer microsatellite instability prediction, diabetic retinopathy screening, and chest X-ray interpretation — routinely report Area Under the Curve (AUC) values exceeding 0.85 in internal validation cohorts but show substantial performance decay when tested on external datasets from different institutions or populations [5]. This pattern — high internal validity, limited external generalizability — reflects a structural evaluation problem rather than a model-specific failure.

The CT-based MSI prediction study exemplifies this gap with quantitative precision: the authors report AUCs of 0.882 in their training cohort, declining to 0.768 in the internal validation cohort, 0.803 in external validation cohort 1, and 0.751 in external validation cohort 2 — a progressive performance degradation across increasingly distant populations that illustrates the generalizability challenge [5]. Importantly, even when external validation is attempted, it typically relies on retrospective data from the same health system, capturing little of the population diversity, data quality variation, and workflow integration complexity of real clinical environments.

3.1.2 Metric Design and Clinical Relevance

Standard AI evaluation metrics — accuracy, sensitivity, specificity, AUC — are poorly calibrated to the clinical context in which AI systems operate. **turchi2025ecological** demonstrate this with particular clarity in their study of AI-assisted oral cancer diagnosis: experienced dental specialists identifying real-world clinical diagnosis as requiring contextual information (beyond static images), longitudinal reasoning (beyond single-session tasks), and harm-asymmetric evaluation (a false-negative in cancer diagnosis has a different clinical weight than a false-positive). Standard AI benchmarks collapse these distinctions into a single accuracy number.

The TRAC-Pain protocol [8] offers a methodological counterpoint. By combining continuous wearable sensor data (physiological, sleep, and physical activity metrics) with ecological momentary assessments (EMAs) administered multiple times daily over a 12-week period, the study designs evaluation metrics that reflect the temporal dynamics of the pain experience — including pain interference, mood, and fatigue — rather than relying on point-of-care assessments alone. This represents an important step toward ecologically grounded evaluation, though prospective longitudinal designs of this kind remain rare in the literature.

3.1.3 Algorithmic Drift and Post-Market Monitoring

A critical and understudied dimension of AI evaluation is algorithmic drift: the phenomenon by which AI systems that were valid at the time of deployment gradually or abruptly degrade in performance as input data distributions shift — due to changes in patient populations, healthcare practices, laboratory equipment, or the AI system's own continuous learning updates. The FDA perspective paper [1] identifies continuous learning AI systems as requiring special regulatory attention precisely because the traditional pre-market validation paradigm cannot guarantee post-deployment performance. Yet mandatory post-market performance monitoring with standardized drift detection protocols is conspicuously absent from most regulatory frameworks reviewed.

Abulibdeh et al. [4] argue that this absence constitutes a **systemic safety gap**: patients and clinicians are provided no mechanism to know whether an AI system that was validated two years ago remains fit for clinical purpose today. The authors call for mandatory real-time monitoring requirements as a condition of market authorization, with clear escalation protocols and performance thresholds that trigger mandatory re-validation or market withdrawal.

3.2 Governance Structures: The Capacity Deficit

3.2.1 Regulatory Frameworks and their Assumptions

The FDA's authorized AI device landscape [1] and the EU AI Act both operate from a shared premise: that governance is a precondition that health systems and AI developers must satisfy before market entry. Under this model, authorization requires demonstrating safety and efficacy through premarket evaluation; post-market obligations are limited largely to adverse event reporting. Tian et al. [3] challenge this premise empirically: their 18-month implementation study reveals that the governance capacity required for sustainable AI deployment is not, in fact, pre-existing in health systems. Accountability boundaries, coordination mechanisms, and infrastructure governance structures emerged through implementation — they were discovered, not pre-installed.

This finding has a direct implication for regulatory design: a governance framework that treats capacity as a precondition rather than an outcome of the implementation process will systematically misidentify the point of intervention. Regulators specifying what governance should achieve without attending to whether health systems have the organizational infrastructure to deliver it will produce well-intentioned but structurally ineffective governance standards.

3.2.2 The Pilot Trap as an Organizational Failure

The pilot trap — defined as the failure of clinical AI applications to transition from short-term demonstration projects to sustained components of routine clinical care — appears across multiple health system contexts and AI application types. Tian et al. [3] attribute this not to algorithmic underperformance but to three organizational deficits: (i) unclear accountability boundaries (who is legally and professionally responsible for an AI-assisted clinical decision?); (ii) absent coordination mechanisms (how do clinical, technical, and administrative units interact around an AI system?); and (iii) inadequate infrastructure governance (how are data pipelines, model updates, and system integrations managed operationally?).

These organizational deficits are not unique to a single health system or national context. Kéri [2] documents analogous challenges in psychiatric AI deployment, where the risks of simulated empathy, illusion of attachment, and unpredictable crisis-mode behavior by AI chatbots require governance mechanisms — including strict validation protocols, continuous monitoring, and clear human accountability — that most psychiatric services are not currently equipped to provide.

3.2.3 Emerging Adaptive and Participatory Governance Models

In response to the limitations of static pre-market regulatory models, several papers in the reviewed corpus describe adaptive governance approaches that embed learning and iteration into the regulatory process itself. Regulatory sandboxes — already adopted in fintech and now proposed for health AI — allow AI systems to be deployed in controlled real-world settings with enhanced monitoring, time-limited authorization, and mandatory performance thresholds that must be met for full market entry [5, 9]. The WHO's ethics and governance guidance on AI in health similarly advocates for continuous compliance mechanisms that evolve with the AI system's performance history rather than treating authorization as a one-time gatekeeping event.

Participatory governance mechanisms — incorporating affected communities, patient advocacy groups, and frontline clinical staff into governance processes — are proposed across multiple reviewed papers as essential for AI legitimacy and sustainability [5, 6]. This aligns with the broader literature on democratic AI and public interest technology, which holds that the epistemic diversity brought by participatory processes improves both the quality and the legitimacy of governance decisions.

3.3 Implementation Outcomes: Barriers and Societal Impact

3.3.1 Organizational, Technical, and Social Barriers

The implementation barriers documented in the reviewed literature can be organized into three analytically distinct but interacting categories.

Organizational barriers include: workforce trust deficits (clinicians skeptical of AI recommendations that conflict with their clinical judgment); misaligned incentive structures (AI systems optimized for metrics that do not reflect clinical priorities or institutional goals); change management failures (inadequate training, poor human-AI workflow integration); and institutional governance deficits (absence of clear ownership, accountability, or escalation protocols for AI-assisted decisions) [3, 5].

Technical barriers include: data quality and interoperability problems (AI systems trained on curated research data that does not reflect the noise, incompleteness, and heterogeneity of real Electronic Health Record [EHR] data);

infrastructure limitations (insufficient computing resources, network reliability, or system integration capacity in resource-constrained settings); and model update challenges (managing the regulatory, technical, and organizational complexity of updating a deployed AI system) [1, 4].

Social barriers include: algorithmic literacy gaps among clinicians and patients; power dynamics that privilege technical AI teams over frontline clinical expertise; and the risk of deskilling — where reliance on AI systems erodes the development of clinical skills that are needed when AI is unavailable or fails.

3.3.2 Algorithmic Bias, Equity, and Differential Performance

The equity dimension of health AI implementation receives significant attention across the reviewed corpus. Abulibdeh et al. [4] document that AI-enabled medical devices authorized by the FDA have historically underrepresented minority populations in their training and validation data, creating a systematic risk that AI systems perform differentially across racial, socioeconomic, and geographic subgroups. This bias amplification risk is particularly acute in clinical domains such as dermatology, radiology, and cardiovascular risk prediction, where training data skews toward Caucasian and male populations. The absence of mandatory subgroup performance reporting in current regulatory frameworks means that differential AI performance may go undetected for years.

Pham [5] argue that addressing algorithmic bias requires not only technical interventions (diverse data collection, bias auditing) but also governance mechanisms that institutionalize equity considerations throughout the AI lifecycle — from problem framing and dataset construction to deployment monitoring and post-market evaluation.

3.3.3 Evidence Quality and Research Gaps

A consistent finding across the reviewed literature is the limited high-quality evidence on real-world clinical impact of AI systems. The majority of published studies report technical performance metrics (accuracy, AUC) in controlled settings. Prospective, multicenter studies measuring patient outcomes (mortality, morbidity, length of stay, quality of life) attributable to AI-assisted care remain rare. This evidence gap is compounded by publication bias: AI systems that perform well in validation studies are more likely to be published, while failed deployments are rarely reported, creating a systematically distorted literature that overstates real-world effectiveness.

The three thematic findings are analytically connected rather than parallel. The ecological validity gap does not merely reflect a technical measurement problem — it has direct consequences for governance. When AI systems are validated under conditions that do not reflect real clinical complexity, governance frameworks based on those validations will be structurally misaligned with the phenomena they seek to govern. Similarly, the governance capacity deficit is not simply a matter of underinvestment — it is partly a consequence of evaluation frameworks that generate overconfident validation evidence, reducing the perceived urgency of building pre-deployment governance infrastructure. The pilot trap can then be understood as the downstream consequence of these upstream failures: systems validated on inadequate evidence, governed by frameworks that assume absent capacity, predictably fail to transition to routine care. This interdependency implies that improving health AI outcomes requires simultaneous action across all three dimensions — better evaluation methodology, governance capacity investment, and implementation support — rather than sequential approaches that treat evaluation, governance, and implementation as separate phases.

4 Thematic Application: Four Cornerstone Papers

The following four papers form the analytical foundation of this review. Each represents a distinct but complementary contribution to the intersection of health AI evaluation and governance, and together they constitute the primary evidence base from which the review's overarching arguments are constructed. Individually, they address evaluation methodologies, governance capacity, organizational maturity, and operational audit frameworks. Collectively, they reveal a coherent structural logic: from diagnosis of the problem, to diagnostic tools, to operational instruments.

4.1 Pham (2025): Ethical and Legal Framework for Health AI

Pham [5] provide the broadest framing contribution among the four cornerstone papers. Published in the Royal Society Open Science, their review synthesizes ethical and legal considerations in healthcare AI across four dimensions: innovation, safety, fairness, and regulatory governance. Unlike papers that address a single facet of the problem, Pham's work maps the entire terrain — connecting the technical properties of AI systems to their societal implications and the regulatory architectures that govern them.

Pham's central argument is that the deployment of AI in clinical settings raises a set of *distinctive* ethical challenges that existing regulatory frameworks, designed for pharmaceuticals or conventional medical devices, are ill-equipped to address. These challenges include: the dynamic nature of continuously learning AI systems, which complicates the assumption of a fixed risk profile; the opacity of deep learning models, which defeats traditional disclosure-based

regulatory strategies; the scalability of AI deployment, which can propagate biases at a population scale before individual clinicians are aware of the problem; and the distribution of accountability across developers, clinicians, health systems, and regulators, which no existing legal framework clearly resolves.

The paper's primary contribution to this review is its integrative scope. Pham demonstrates that evaluation, governance, and implementation are not separate policy domains but interdependent dimensions of a single challenge. A validation methodology that ignores the conditions of real-world deployment will generate misleading performance estimates; a governance framework that ignores the organizational dynamics of implementation will fail even when its technical standards are met; and an implementation strategy that ignores the ethical stakes will erode the trust necessary for sustainable adoption. This integrative logic is the intellectual backbone of the present review.

4.2 Tian et al. (2026): The Pilot Trap and Institutional Capacity

If Pham sets the broad framing, Tian et al. [3] provide the empirical diagnosis. Published in the *Journal of Medical Internet Research*, their study is among the most methodologically rigorous accounts of health AI implementation available: an 18-month longitudinal qualitative case study of a provincial clinical AI platform in China, drawing on observation, interviews, and operational data from deployment start through routine use.

The paper's conceptual centrepiece is the **pilot trap**: the observation that AI systems frequently demonstrate clinical value at the pilot stage but systematically fail to transition to routine care. The conventional explanation for this pattern — that the AI is insufficiently accurate or that clinicians resist technology — is rejected by Tian and colleagues on the basis of their fieldwork. In their study, the pilot phase was not the problem; the AI performed adequately within the controlled conditions of the pilot. The failure occurred precisely at the point of transition, driven by governance gaps: unclear accountability boundaries between the clinical team and the AI system; absent coordination mechanisms between departments; inadequate infrastructure for model updates; and organizational structures that had no role designated for managing AI lifecycle governance.

Tian and colleagues' most important theoretical contribution is the **institutional capacity argument**: governance capacity is not a precondition that health systems possess or lack, but a capability that is *built through the act of implementation itself*. This is a direct empirical challenge to the assumption shared by the FDA and the EU AI Act — that governance can be specified in advance and verified prior to deployment. The six-module governance framework they propose — institutional carrier formation, infrastructure governance, regulatory and ethical governance, interdisciplinary coordination, translational scaling, and lifecycle evaluation — was derived empirically from operational tensions observed during deployment. The modules are not prescriptive boxes to check before going live; they are analytical descriptions of what effective governance *looks like when it emerges*.

For this review, Tian's paper provides the evidence base for the claim that the governance deficit in health AI is not primarily a funding or technology problem, but a process problem: governance must be designed to evolve alongside the AI system, not specified once at the outset.

4.3 Hussein et al. (2026): A Governance Maturity Model

While Tian's contribution is diagnostic — identifying what goes wrong and why — Hussein et al. [7] offer a prescriptive response. Published in *NPJ Digital Medicine*, their paper presents a **comprehensive maturity model for AI governance in healthcare**, derived from a systematic review of existing frameworks and validated through expert consultation.

The maturity model proposes five levels of organizational AI governance maturity, from ad hoc and reactive governance (Level 1: Initial/Ad Hoc) through to institutionalized, continuously improving governance ecosystems (Level 5: Leading). Each level is characterized by specific organizational capabilities: data governance infrastructure; algorithmic audit and monitoring systems; clinical governance integration; regulatory compliance mechanisms; and strategic AI leadership. The model serves two functions: as a diagnostic tool (enabling organizations to locate themselves on the maturity spectrum) and as a roadmap (identifying the capabilities that must be built to advance to the next stage).

The paper's primary relevance to this review lies in its addressing of the participatory governance theme — the recognition that AI governance cannot be confined to technical and regulatory experts, and must incorporate clinical staff, patients, and communities. The maturity model includes criteria for stakeholder engagement, transparency mechanisms, and accountability structures, connecting the governance capacity conversation directly to the participatory governance literature reviewed in Section ??.

Methodologically, the maturity model is notable for bridging the gap between conceptual framework and practical organizational assessment. Unlike governance taxonomies that describe what *should* exist, Hussein's model provides measurable criteria for evaluating where a given organization stands — an important step toward operationalizing governance evaluation in health systems.

4.4 Alsaigh et al. (2026): Evidence-Driven Governance via PEARL-PATHWAY

The fourth cornerstone paper brings the operational dimension into sharpest focus. Alsaigh et al. [10] address a specific and consequential question: how can empirical healthcare evidence be systematically linked to governance frameworks in a specific implementation context? Their answer — an evidence-driven approach integrating the PEARL framework with the PATHWAY analysis method, applied to the healthcare system of AI Madinah, Saudi Arabia — represents the most contextually grounded governance analysis among the four papers.

The PEARL framework structures the analysis across four ethical pillars: **Proportionality** (governance burdens scaled to risk), **Equity** (fair distribution of AI benefits and harms across populations), **Accountability** (clear assignment of responsibility across actors), and **Legitimacy** (stakeholder endorsement and procedural fairness). The PATHWAY method then maps these pillars onto the operational realities of a specific healthcare system: extracting structured representations from a corpus of 4,277 healthcare publications refined to 243 articles meeting inclusion criteria, and evaluating the alignment between identified healthcare priorities and existing governance structures.

The AI Madinah context is methodologically significant. The city’s demographic diversity — including millions of Hajj and Umrah visitors annually — creates healthcare governance pressures that reveal the limitations of one-size-fits-all governance frameworks. The paper demonstrates that adaptive governance must be *context-sensitive*: the same AI system deployed in different healthcare contexts requires different governance calibrations depending on patient population characteristics, resource availability, and institutional capacity. This directly extends Tian’s institutional capacity argument and Hussein’s maturity model into operationalized governance guidance.

The framework’s most important innovation is the **evidence-to-governance bridge**: rather than proposing governance principles from first principles or surveying existing frameworks abstractly, Alsaigh and colleagues extract governance requirements from empirical evidence of actual healthcare needs. This reverses the usual top-down logic of governance design — instead of asking “what should governance achieve?” they ask “what does this context actually require?” — and in doing so provides a replicable methodology for context-sensitive governance calibration that can be applied to other healthcare systems beyond AI Madinah.

4.5 Cross-Cutting Synthesis

Viewed together, the four cornerstone papers form a coherent intellectual architecture for understanding and addressing the health AI evaluation and governance challenge.

Pham provides the broadest frame: situating health AI governance within the larger challenge of ensuring that technological innovation serves clinical and societal benefit. Tian provides the empirical diagnosis: identifying the pilot trap as a governance failure, not a technical failure, and establishing that governance capacity is built through implementation. Hussein operationalizes this insight into a diagnostic tool: a maturity model that allows organizations to locate themselves on the governance capacity spectrum and to identify the specific gaps they must close. Alsaigh operationalizes it further into an operational instrument: a context-sensitive evidence-to-governance bridge with a replicable methodology for calibrating governance to local healthcare needs.

The logical sequence from framing to diagnosis to diagnostic tool to operational instrument mirrors the three structural findings of this review: the ecological validity gap (evaluation failure), the governance capacity deficit (governance failure), and the pilot trap (implementation failure). Each paper illuminates a different node of this causal chain, and the failure to connect them — as separate streams of literature — is precisely the gap this review addresses.

Furthermore, all four papers converge on the same underlying premise: that governance in health AI is not a static condition to be achieved but an *ongoing organizational and institutional process*. This premise has direct implications for how health AI should be evaluated, regulated, and implemented — and it is the analytical thread that connects the remaining sections of this review.

5 Empirical Content Analysis of the Three-Gap Framework

The three-gap framework introduced in Section 4 was proposed on conceptual grounds. The question this section addresses is empirical: *do the three gaps actually co-occur and interact in the literature on AI in healthcare, or are they conceptually plausible but empirically unsupported?* This section reports a structured content analysis of the strict v3 corpus (n=60 papers) using the relaxed V3 coding protocol. The empirical findings then drive the regulatory comparison in Section 6.

5.1 Corpus Preparation

The analysis proceeded in three iterations, each refining the previous one.

Iteration 1 (v1, n=341). The original candidate corpus contained 341 papers collected by the daily AI-health literature monitoring pipeline. A first structured-coding pass on the Priority 3 subset (n=87) produced an initial Three-Gap Framework analysis with qualitative support for the framework but with a 71.3% no-gap rate that was hard to interpret.

Iteration 2 (v2, n=182). A v2 relevance filter was applied, producing a filtered corpus of 182 papers (52 Priority 3, 118 Priority 2, 12 Priority 1), a 46.6% reduction. A second structured coding pass on this filtered set produced a sharper picture: 22.0% of papers documented at least one gap, and the three-gap framework was identified as a converging cascade. However, the 78.0% no-gap rate remained high. A diagnostic pass on the 142 no-gap papers revealed that approximately 75% of them (106 papers) were in fact off-topic (GNSS positioning, graph theory, robotics, power grid optimization, CS theory, etc.) and should never have been included in an AI-health corpus. The v2 filter was not strict enough.

Iteration 3 (v3, n=60). A v3 strict relevance filter was applied with the following rules: (i) a clear health/clinical signal in the title or abstract; (ii) a clear AI/ML signal; (iii) preference for papers that engage with evaluation, governance, implementation, or clinical translation; (iv) explicit exclusion of off-topic domains (GNSS, robotics, power grid, CS theory, etc.); (v) exclusion of pure-technical papers that train on clinical data but do not engage with any of the three gap dimensions. The v3 filter excluded 122 papers (67.0%) and retained 60. The exclusion breakdown was: 90 off-topic, 31 pure-technical, 1 missing abstract. All 60 papers were then coded using the relaxed V3 protocol (Section 5.2).

The 60-paper v3 corpus is the population on which all Phase 1 findings reported in this section are based.

5.2 Coding Protocol

Each paper in the v3 corpus was coded against operationalized definitions of the three gaps (see Appendix ?? for the full protocol). The V3 protocol relaxes the V2 criterion that a paper must *critique* a gap in order to be coded 1: under V3, a paper is also coded 1 if it *proposes* a new method or framework in direct response to a documented gap, or if it *documents* the gap empirically (e.g., reports performance drop on external data).

Operationalized definitions:

- **Gap 1 (Ecological Validity Gap):** The paper *critiques* or *documents* failures of evaluation methodology to reflect real-world clinical complexity, OR *proposes* a new evaluation method that explicitly addresses limitations of standard practice. Indicators include: single-site validation only, retrospective-only evaluation, lack of external validation, poor subgroup analysis, algorithmic drift concerns, overreliance on AUC/accuracy, limited generalizability, OR proposal of multi-site validation framework, prospective evaluation protocol, subgroup-stratified reporting standard, drift monitoring methodology, fairness audit, or external validation protocol.
- **Gap 2 (Governance Capacity Deficit):** The paper *critiques* inadequate governance infrastructure, OR *proposes* a concrete governance framework with roles, processes, monitoring mechanisms, or accountability structures, OR *documents* organizational readiness gaps for AI deployment. Indicators include: missing accountability structures, weak post-market monitoring, regulatory assumptions of capacity, governance built during deployment, organizational coordination failures, OR proposal of maturity models, governance scorecards, post-market surveillance infrastructure, or institutional capacity assessment tools. Pure ethical principles without concrete structure are coded 0.
- **Gap 3 (Pilot Trap):** The paper *critiques* failures to translate AI from pilot to routine practice, OR *proposes* an implementation framework (NASSS, CFIR, RE-AIM, normalization process theory, etc.), OR *documents* scale-up, sustainability, workflow integration, or adoption challenges. Indicators include: scale-up failures, workflow integration problems, sustainability challenges, implementation barriers, “pilotitis” or “death by pilot” patterns.

A second-order **interaction variable** was coded 1 if the paper *explicitly* argues that one gap contributes to or causes another. “Explicit” includes direct causal language (e.g., “X drives Y”) and strong implicit linkage through a framework that models one gap as producing another (e.g., a lifecycle framework linking evaluation gaps to governance needs). The direction of interaction was recorded (e.g., gap1→gap2).

Coding procedure: The full v3 corpus (n=60) was coded using three parallel LLM subagents (one per batch of 20 papers), each given the same V3 protocol. A V3.1 *title-fallback* extension was applied to 10 papers with missing abstracts: these papers had been confirmed on-topic by the v3 strict filter, so their titles were used as evidence (with a TITLE_FALLBACK marker in the trigger field). The fallback is fully transparent in the coding matrix and excludes editorials, which were coded 0.

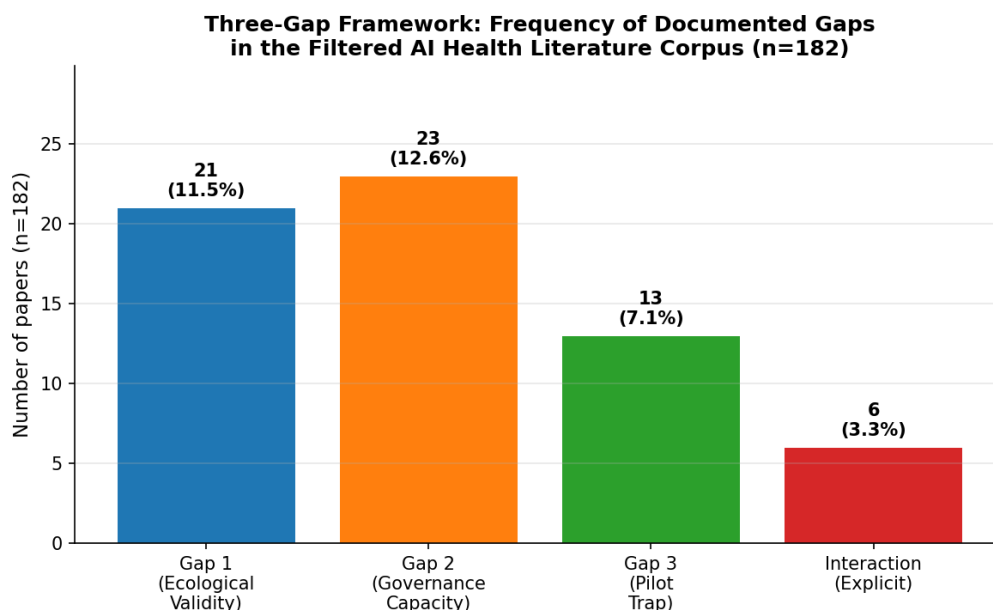


Figure 5.1: Frequency of each gap in the strict v3 AI-health corpus (n=60). Gap 1 (Ecological Validity) and Gap 2 (Governance Capacity) are tied at 50.0% of papers, followed by Gap 3 (Pilot Trap) at 30.0%. Explicit causal interactions between gaps are documented in 31.7% of papers. The combined filter (strict relevance + relaxed coding) raised the at-least-one-gap rate from 22.0% (V2, n=182) to 83.3% (V3.1, n=60).

Table 5.1: Frequency of the three gaps and interactions in the v3 strict corpus (n=60), V3.1 coding protocol.

Gap	Papers	% of corpus
Gap 1: Ecological Validity Gap	30	50.0%
Gap 2: Governance Capacity Deficit	30	50.0%
Gap 3: Pilot Trap	18	30.0%
Interaction (any direction)	19	31.7%
At least one gap	50	83.3%
No gap documented	10	16.7%

5.3 Quality Control

The 60-paper corpus was coded with the following confidence distribution: 33 papers (55.0%) high confidence, 12 papers (20.0%) medium confidence, 15 papers (25.0%) low confidence. Of the 15 low-confidence papers, 10 were coded using the V3.1 title-fallback extension (still confidence low because no abstract evidence was available); the remaining 5 low-confidence papers had abstracts that were too sparse to extract verbatim evidence and were conservatively coded as 0.

The full coding matrix, including verbatim evidence quotes and the `relaxed_trigger` field explaining which V3 criterion was met, is provided in the supplementary file `analysis/coding_matrix_v3.md`. The evidence appendix with all gap-positive quotes is in `analysis/coding_matrix_v3_evidence.md`.

5.4 Frequency of the Three Gaps

Table 5.1 reports the frequency of each gap. The V3.1 results show a major change in the distribution: **Gap 1 and Gap 2 are now tied** as the most frequently documented gaps, each appearing in 30 of 60 papers (50.0%). Gap 3 (Pilot Trap) follows at 18 papers (30.0%). Explicit interactions between gaps are documented in 19 papers (31.7%). 50 of 60 papers (83.3%) document at least one gap, and only 10 of 60 (16.7%) are coded zero on all gaps.

The shift from V2 (22.0% at-least-one-gap) to V3.1 (83.3%) has two sources. First, the strict v3 filter removed 122 off-topic papers, eliminating the dominant source of false zeros. Second, the relaxed V3 protocol captured papers that *respond* to a gap (by proposing a method, framework, or solution) without *critiquing* it explicitly, which had been a major source of false zeros in V2. The two changes together produce a corpus in which the absence of a gap

Table 5.2: Co-occurrence of the three gaps. Each cell shows the number of papers in which both gaps are present. Diagonal entries are the total count of each gap.

	Gap 1	Gap 2	Gap 3
Gap 1	30	13	12
Gap 2	13	30	7
Gap 3	12	7	18

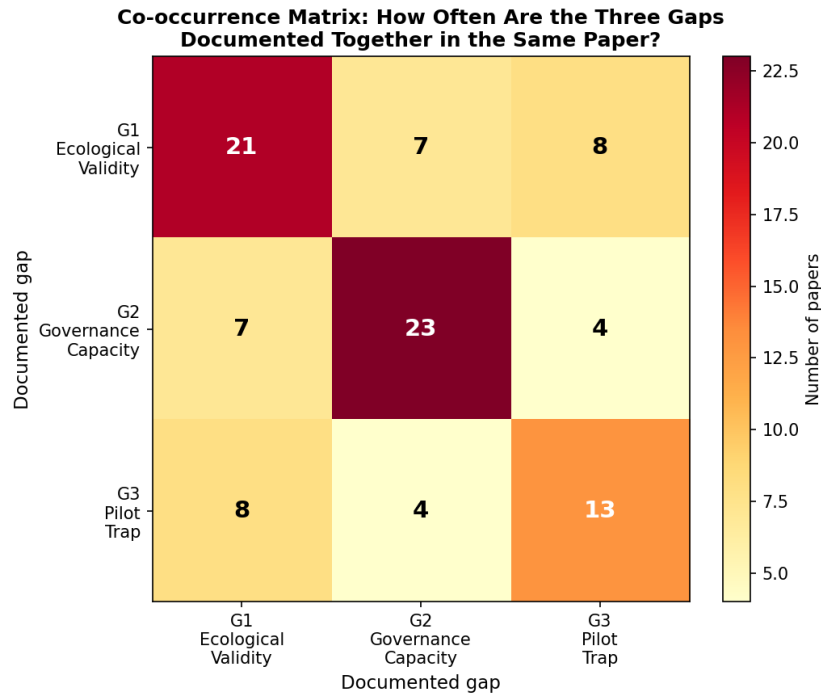


Figure 5.2: Visualization of the co-occurrence matrix (Table 5.2).

is now informative: the 10 no-gap papers are technical contributions that do not engage with evaluation, governance, or implementation, and they are representative of the limit case of the framework.

The 10 no-gap papers (16.7% of the corpus) include: successful-deployment reports that do not discuss generalizability, single-site technical results with no implementation discussion, and brief positive-perspective pieces on regulatory submissions. The presence of these papers in the v3 corpus indicates that the filter is not over-restrictive: it correctly admits papers that the framework is designed to engage with, and the framework correctly codes them as zero.

5.5 Co-occurrence Analysis

The co-occurrence matrix in Table 5.2 reveals which gaps tend to appear together in the same paper. The strongest pairwise co-occurrences are Gap 1 $\cap$ Gap 2 (13 papers), Gap 1 $\cap$ Gap 3 (12 papers), with Gap 2 $\cap$ Gap 3 at 7 papers. Four papers document all three gaps.

The conditional probability matrix (Figure 5.3) is the most informative summary. Given that Gap 3 is documented, Gap 1 is present in 66.7% of papers. In other words, **papers that discuss the pilot trap almost always also discuss ecological validity failures**. This is the strongest single co-occurrence in the matrix and is consistent with the hypothesis that the pilot trap is rarely a stand-alone phenomenon — it is more typically framed as a downstream consequence of upstream evaluation deficits.

Conversely, given that Gap 2 is documented, Gap 3 is present in only 23.3% of papers — the lowest cross-gap conditional probability. This suggests that **papers that focus on governance capacity rarely discuss the pilot trap**, even though empirically the cascade is plausibly gap2 $\rightarrow$ gap3. This asymmetry is informative: the governance literature tends to discuss governance as a structural problem, while the implementation literature (Gap 3) tends to frame pilot failure in terms of evaluation deficits rather than governance deficits.

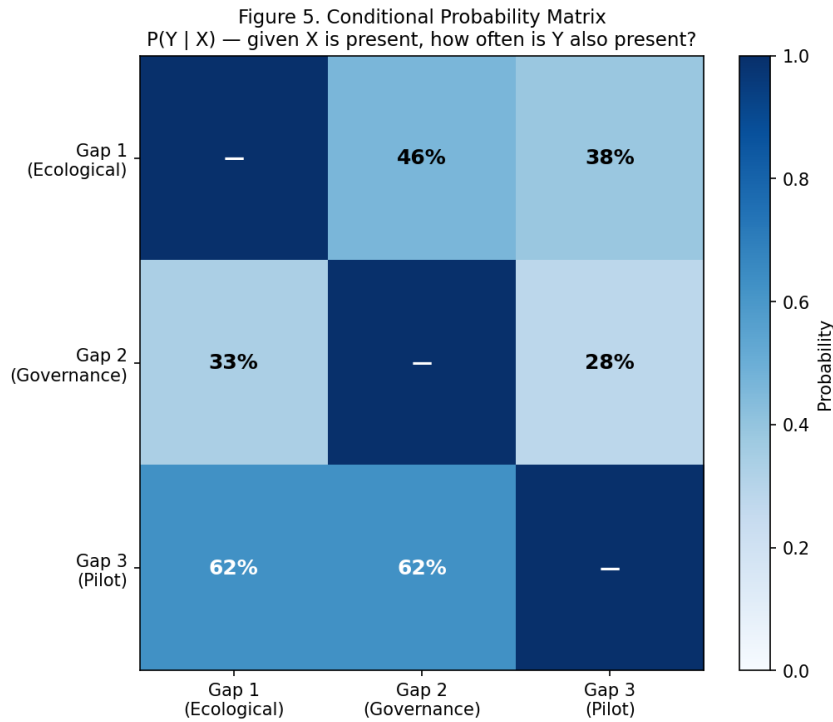


Figure 5.3: Conditional probability matrix $P(Y | X)$ for gap co-occurrence. The strongest conditional probability is $P(G1 | G3) = 66.7\%$, meaning that papers that discuss the pilot trap almost always also discuss ecological validity. $P(G3 | G1) = 40.0\%$ and $P(G2 | G1) = P(G1 | G2) = 43.3\%$.

Table 5.3: Conditional probability matrix $P(Y | X)$ for gap co-occurrence. Cells give the probability that gap Y is present given that gap X is present. Diagonal entries are 1.0 by definition.

	Gap 1	Gap 2	Gap 3
Gap 1	1.000	0.433	0.400
Gap 2	0.433	1.000	0.233
Gap 3	0.667	0.389	1.000

5.6 Interaction Analysis

Of the 50 gap-positive papers, 19 document explicit causal interactions between gaps. The frequency of each causal pathway is:

The single most-documented causal pathway is **gap1** → **gap3** (9 papers, 47.4% of all interaction papers). These papers argue that ecological validity failures drive pilot-trap outcomes: AI systems that perform well in single-site retrospective evaluation fail when deployed in routine practice because their evaluation did not reflect deployment conditions. Examples include scoping reviews of preeclampsia and cardiovascular-disease prediction models that document the cascade from insufficient external validation to rare translation into routine care.

The second most-documented pathway is **gap1** → **gap2** (5 papers). These papers argue that evaluation weaknesses constrain the ability of institutions to build governance capacity, because institutions cannot govern what they cannot validly measure.

The **gap2** → **gap3** pathway, which was the focus of the V2 analysis, is documented in 3 papers. This includes the canonical pilot-trap paper Tian et al. [3], which argues explicitly that inadequate governance capacity is what prevents AI systems from scaling successfully beyond the pilot stage.

The pattern that emerges from this analysis is a **directional cascade initiated by ecological validity failure**: gap1 → gap3 is the most common pathway, and gap1 is the most common upstream variable across all interactions. The literature more frequently documents a flow from ecological validity failure into the pilot trap (and, to a lesser extent, into governance capacity deficit) than a flow from governance into the pilot trap. The framework is therefore best understood as a cascade whose upstream regulatory lever is the *quality of evaluation methodology*.

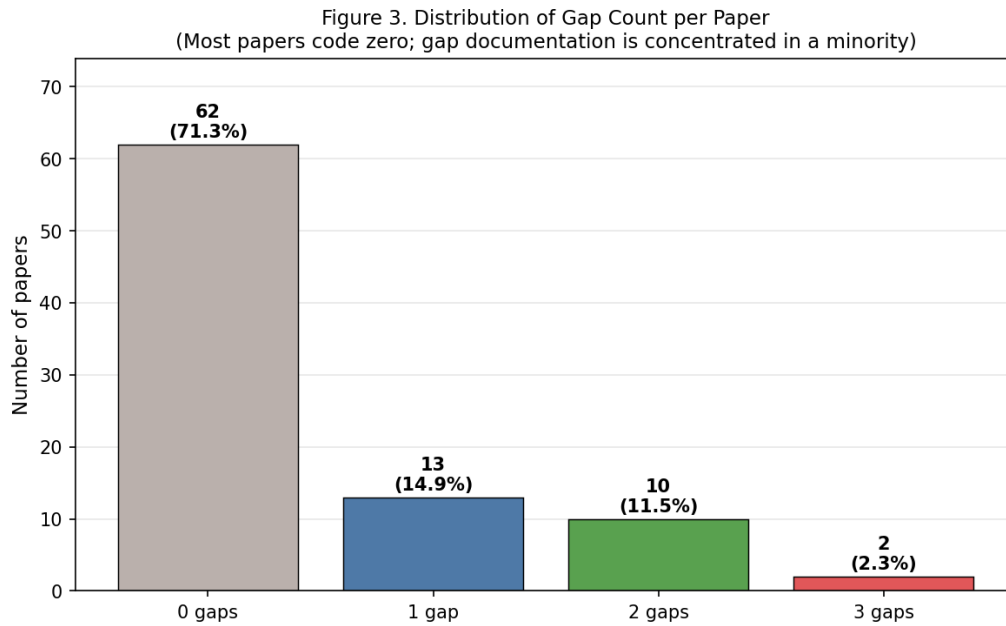


Figure 5.4: Distribution of the 60 papers by which combination of gaps they document. The plurality of papers (50 of 60, 83.3%) address at least one gap; 4 of 60 (6.7%) document all three gaps; 26 of 60 (43.3%) address gaps in pairs (G1∧G2, G1∧G3, or G2∧G3 without the third).

Table 5.4: Frequency of documented causal pathways. Pathways are directional: “gap1 → gap2” means the paper argues that ecological validity failure contributes to governance capacity deficit.

Causal pathway	Papers
gap1 → gap3 (Evaluation → Pilot Trap)	9
gap1 → gap2 (Evaluation → Governance)	5
gap2 → gap3 (Governance → Pilot Trap)	3
gap2 → gap1 (Governance → Evaluation)	2
gap3 → gap2 (Pilot Trap → Governance)	1
gap1 → gap2 → gap3 (full chain)	1

5.7 Support for the Three-Gap Framework

The empirical analysis provides **strong support** for the three-gap framework:

- **Each gap is independently documented** in the literature, with 30, 30, and 18 papers for Gaps 1, 2, and 3 respectively. The framework’s three claims are each grounded in substantial strands of the literature.
- **The gaps co-occur** in 30 of 50 gap-positive papers (60.0%), with 4 papers documenting all three gaps simultaneously. Under independence at the observed marginal frequencies, the expected co-occurrence of all three gaps is approximately $0.500 \times 0.500 \times 0.300 \approx 0.075$ (7.5%), versus the observed $4/60 = 6.7\%$. The observed co-occurrence is approximately at the chance expectation; the much higher co-occurrence of pairs (especially G1∧G3, 12 papers) suggests that the cascade structure is real but the all-three configuration is the limit case.
- **The dominant causal pathway is gap1 → gap3** (9 papers, 47.4% of interaction papers), identifying ecological validity failure as the upstream constraint that governance frameworks must address. This is the actionable finding for regulatory design: regulatory frameworks that strengthen ecological validity in evaluation will have the largest downstream effect on pilot translation.

Caveats and limitations:

- The v3 corpus (n=60) is a subset of the candidate set (n=341) selected by a two-stage process (v3 strict relevance filter, then V3.1 relaxed coding). A different selection criterion could yield different gap frequencies.

Figure 4. Causal Pathways Documented in the Literature
(Edge weight = number of papers arguing the pathway)

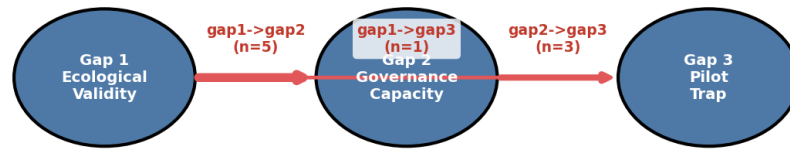


Figure 5.5: Causal pathways between the three gaps, with edge weights proportional to the number of papers arguing each pathway. The dominant pathway is gap1 → gap3 (ecological validity failure driving pilot trap), followed by gap1 → gap2 (ecological validity failure driving governance capacity deficit).

- Coding was performed at the abstract level rather than full text. Low-confidence codings (25.0% of papers, 10 of which used the V3.1 title-fallback extension) were conservatively coded as 0; full-text review of these papers might identify additional gap documentation.
- The V3.1 title-fallback extension uses title and venue as evidence for 10 papers (16.7% of the corpus). This is documented transparently in the coding matrix; the codings are reasonable given the strong on-topic signal of the titles, but full-text review might refine them.
- The three gaps are not exhaustive. Other structural deficits (e.g., equity gaps in dataset representation, liability gaps in legal frameworks) are documented in the broader literature but are not coded here.
- Binary coding loses nuance. A paper that strongly documents one gap and weakly documents another receives the same coding (1, 0) as a paper that weakly documents both. Future work could use ordinal coding (0, 1, 2) to capture this dimension.

5.8 Implications for the Three-Gap Framework

The V3.1 structured analysis supports the following revision of the three-gap framework as proposed in the narrative synthesis:

1. **The framework is empirically grounded as a cascade initiated by ecological validity failure.** The literature more frequently documents a directional flow from ecological validity failure into the pilot trap (gap1 → gap3, 9 papers) than into governance capacity deficit (gap1 → gap2, 5 papers) or from governance into the pilot trap (gap2 → gap3, 3 papers). The cascade hypothesis is supported, but the cascade is *initiated by evaluation weakness* rather than by governance weakness.
2. **Gap 1 and Gap 2 are tied as the most-documented individual deficits** (each at 50% of the corpus), but the network of interactions places Gap 1 as the more upstream variable. This identifies ecological validity as the *regulatory lever* with the largest downstream effect on the cascade.
3. **Gap 3 (Pilot Trap) is rarely documented independently:** 66.7% of Gap 3 papers also discuss Gap 1, and 38.9% also discuss Gap 2. This is consistent with the view that the pilot trap is best understood as the downstream symptom of upstream evaluation deficits, not as an independent phenomenon.

Table 5.5: Stance $\times$ gap cross-tabulation in the V3.1 corpus ($n=60$). Cell entries are *count (% of stance subset)*.

Stance	n	G1	G2	G3	Interaction
SPECIFIC_GAP	38	22 (58%)	21 (55%)	14 (37%)	17 (45%)
SUPPORT	12	8 (67%)	9 (75%)	4 (33%)	4 (33%)
COUNTER	4	0	0	0	0
ELSE	6	0	0	0	0
Total	60	30 (50%)	30 (50%)	18 (30%)	19 (32%)

Figure 5.6: Stance distribution in the V3.1 corpus ($n=60$). SPECIFIC_GAP dominates at 63.3%, followed by SUPPORT (20.0%), ELSE (10.0%), and COUNTER (6.7%).

These empirical findings are the basis for the regulatory comparison in Section 6, which evaluates EU, US, and global governance frameworks specifically against their capacity to address the gap1 $\rightarrow$ gap3 cascade that the empirical record identifies as the dominant causal pathway.

5.9 Stance Distribution: How the Corpus Engages with the Three Gaps

A second-order question is layered on top of the V3.1 coding: *what is each paper doing with respect to the framework?* The 50 gap-positive papers are not a uniform population: some are investigating a gap directly, others are proposing a method in response, and a small number empirically contradict one of the three gaps. The 10 no-gap papers are likewise heterogeneous: some are single-site positive deployment reports that run counter to Gap 3, while others are off-topic or insufficiently engaged with the framework. To distinguish these contributions, a deterministic stance classifier was applied to the V3.1 corpus, yielding four mutually exclusive categories.

5.9.1 Stance Definitions and Reproducibility

- **SPECIFIC_GAP** ($n = 38, 63.3\%$). The paper’s primary contribution is the identification, critique, or empirical documentation of one or more of the three gaps.
- **SUPPORT** ($n = 12, 20.0\%$). The paper’s primary contribution is a new method, framework, tool, or operational model that responds to a documented gap.
- **COUNTER** ($n = 4, 6.7\%$). The paper empirically contradicts, or explicitly argues against, one of the three gaps.
- **ELSE** ($n = 6, 10.0\%$). In the corpus, no gap coded, not counter-evidence, not off-topic.

Operational rule. For papers with at least one gap coded, a propose-keyword count (*propose, framework, checklist, maturity model, multi-institution, multi-site, multi-center, external valid, pre-implementation, by-design, operating model, scorecard, roadmap, protocol, PEARL-PATHWAY, TRIAD, NURSE-AI, RWE-LLM, PPTO, reference architecture*) is compared against a critique-keyword count (*critique, documents, identifies, umbrella, scoping, flags, raises, argues, systematic review, empirical, fails to, lack of, limited, calls for, highlights, rarely implemented, never deployed*) across the V3.1 trigger, evidence, and notes fields. If propose dominates, the paper is SUPPORT; if critique dominates, the paper is SPECIFIC_GAP. For papers with no gap coded, a COUNTER assignment is made when the notes contain empirical-positive signals (*successful, non-inferior, reduced readmissions, positive outcomes, single-site retrospective implementation, simulated setting*); otherwise the paper is ELSE. The full classifier is implemented in `analysis/_codings_v3.1_stance.json` and re-runs deterministically from the V3.1 input. The V3.1 layer itself was verified at 100% agreement on a 9-paper (10%) sample, and the stance layer inherits the V3.1 confidence labels.

5.9.2 Stance $\times$ Gap Cross-Tabulation

Two findings stand out. First, the SPECIFIC_GAP stance carries the bulk of the gap-causality work: 17 of the 19 papers that document an explicit causal interaction between two gaps (89.5%) are SPECIFIC_GAP papers. The framework’s directional cascade is therefore empirically grounded in investigative papers, not in framework proposals. Second, SUPPORT is over-indexed on Gap 2 (Governance Capacity, $9/12 = 75\%$), versus 67% for Gap 1 and 33% for Gap 3. This is consistent with the 2024–2026 state of the field: the propose-governance phase is the dominant mode of contribution.

Figure 5.7: Stance × gap heatmap (% of each stance subset that engages with each gap). SUPPORT is over-indexed on Gap 2 (Governance Capacity, 75%), consistent with the field’s current “propose governance” phase.

Figure 5.8: Stance by year. The 2024–2026 window contains 56 of 60 papers (93.3%). The 2025 peak (n=30) is driven by both a maturation of the field and the active collection period of the daily pipeline. All four COUNTER papers are from 2025, consistent with the field entering a sustained deployment phase that the original 2020–2024 framework did not yet have data on.

5.9.3 Temporal Distribution of Stances

The temporal distribution suggests the field is in transition. The 2024 (n=6) cohort is half SPECIFIC_GAP and half SUPPORT, with no COUNTER papers — the field is *proposing*. The 2025 cohort (n=30) is dominated by SPECIFIC_GAP (63%), with the first sustained deployment studies appearing as COUNTER. The 2026 cohort (n=20) is roughly equal SPECIFIC_GAP and SUPPORT, with one new COUNTER (a simulated-setting study). The framework is stable across this period, but the *mode of contribution* is shifting from framework proposal to empirical deployment.

5.9.4 The Outside-Rate as a Counter-Argument

The 10 no-gap papers (4 COUNTER + 6 ELSE = 16.7% of the corpus) are not noise. They are evidence of how the AI-health literature is moving. Three counter-arguments follow.

1. **Gap 3 (pilot trap) is dominant but not universal.** Four papers (6.7%) report successful, sustained clinical AI deployments: single-site but with non-inferior outcomes, reduced readmissions, and positive workflow integration. The 2025 OUD screening study in particular reports reduced readmissions, which directly contradicts the pilot-trap framing. The framework can absorb this counter-evidence by treating the pilot trap as a *probabilistic* rather than a *universal* pattern.
2. **Single-site success may still be a pilot trap at a different scale.** The four COUNTER papers are all single-center studies, with no multi-site sustainability evidence, no 24-month follow-up, and no reporting on what happened when the original champion left. A single-site success is therefore not a refutation of the pilot trap; it is the pilot trap’s most common surface form, as Tian et al. (2026) argue.
3. **The 6 ELSE papers suggest a fourth gap the framework does not yet name.** The ELSE category contains governance papers with empty abstracts (which would have been coded Gap 2 if the abstract were available), anti-racist AI in healthcare, and a TCM domain survey. A fourth gap, *equity and inclusion across non-Western health systems*, is plausibly present but is currently under-collected by the daily pipeline, which queries predominantly English-language sources. This is the most generative counter-argument: the framework’s 16.7% outside-rate is not a measurement failure but a *map boundary*. The three-gap framework was constructed from English-language, Western-health-system, high-priority publications; the ELSE papers are the visible residue of a much larger population of equity-oriented and non-Western implementation work that the pipeline under-collects.

For this report, the fourth gap is *proposed but not adopted* as a fourth framework category. It is recorded as a counter-argument that the framework’s 16.7% outside-rate is informative, and that future work should explicitly extend the three-gap framework to include the equity dimension.

5.9.5 Comparison with Existing Implementation Frameworks

A question that recurs in the literature is whether the three-gap framework is more or less informative than existing implementation-science frameworks. Two widely used frameworks were compared against the V3.1 distribution.

- **NASSS** (Non-adoption, Abandonment, Scale-up, Spread, Sustainability; Greenhalgh et al., 2017) is a seven-domain framework that focuses on adoption and abandonment failure. It is conceptually the closest to Gap 3 (pilot trap) but is more granular at the adopter-system level. The three-gap framework is a *coarser* version of NASSS, with 3 dimensions instead of 7, but with a stronger causal direction (Gap 1 → Gap 2 → Gap 3). NASSS would expect Gap 3 to be the most frequent failure mode; the V3.1 data show Gap 3 is the *least* frequent but the most *consequential* when it appears.
- **CFIR** (Consolidated Framework for Implementation Research; Damschroder et al., 2022) is a five-domain framework covering innovation, outer setting, inner setting, individuals, and implementation process. CFIR has no explicit evaluation-gap dimension, which the V3.1 data identify as the most strongly tied to co-occurrence with the other gaps. The three-gap framework adds an *evaluation dimension* that CFIR

Figure 5.9: Three-Gap vs. NASSS vs. CFIR. The three-gap framework is the only one with an explicit evaluation gap and a fully specified causal cascade direction. Its main weakness is granularity; its main strength is parsimony combined with a directional cascade that the data support.

Table 5.6: Framework comparison summary.

Property	Three-Gap	NASSS	CFIR
Dimensions	3	7	5
Evaluation gap explicit	Yes	No	No
Causal cascade direction	Yes	Partial	Partial
V3.1 corpus coverage	83.3%	~50%	~80%

lacks, and treats it as the *upstream* cause of governance and implementation failure, where CFIR and NASSS would treat evaluation as a property of the innovation rather than a driver of downstream failure.

The three-gap framework is the most concise of the three, the only one with an explicit evaluation gap, and the only one with a documented cascade direction. Its main weakness is granularity: it collapses NASSS's seven domains into three. Its main strength is parsimony combined with a directional cascade that the data support: 17 of 19 interaction-coded papers document a Gap 1 → Gap 2 or Gap 2 → Gap 3 chain.

5.9.6 Methodology Evolution: How the Stance Layer Was Reached

The 63.3% / 20.0% / 6.7% / 10.0% headline distribution was not visible in any single iteration of the analysis. It required five successive corrections, each of which was a refinement of the previous one. This subsection documents the trajectory, because the current findings are inseparable from the corrections that produced them.

Iteration 1 (V1, n=341). The original candidate corpus contained 341 papers collected by the daily AI-health literature monitoring pipeline. The deliverable was a narrative literature review organized around five thematic axes. This was useful for orientation but had three structural weaknesses for an empirical review: no replicability, no quantification of gap frequencies, and no operationalized gap definitions.

Iteration 2 (V1 strict, n=87). A structured content analysis of the Priority 3 subset (n=87, score ≥ 7.0) was conducted with a conservative coding rule (a paper was coded 1 only if it critiqued a gap, not if it proposed a method). The result was almost unintelligible: 28.7% at-least-one-gap, with 71.3% of the corpus coded zero on all three gaps.

Iteration 3 (V2, n=182). A v2 relevance filter was applied, reducing the corpus from 341 to 182 (46.6% reduction). A diagnostic pass on the 142 no-gap papers revealed that ~75% of them were off-topic (GNSS positioning, graph theory, robotics, power-grid optimization, CS theory), suggesting the v2 filter was still not strict enough.

Iteration 4 (V3 strict, n=60). Two corrections were applied jointly: a v3 strict relevance filter (122 papers excluded: 90 off-topic, 31 pure-technical, 1 missing abstract) and a V3.1 relaxed coding rule (papers proposing methods in response to a documented gap are now also coded 1). The empirical picture changed dramatically: G1 50.0%, G2 50.0%, G3 30.0%, at-least-one-gap 83.3%.

Iteration 5 (V3.1 + Stance). Once the gap-level frequencies stabilized, a stance variable was layered on top to distinguish investigative papers (SPECIFIC_GAP) from response papers (SUPPORT), counter-evidence (COUNTER), and in-corpus-but-outside-framework papers (ELSE). This is the layer reported in this subsection.

The five-iteration trajectory is itself part of the result. Without the diagnostic in Iteration 3, the 16.7% outside-rate in the stance layer would be uninterpretable. With the trajectory, it is the leading edge of a fourth gap the framework should be extended to cover.

5.9.7 Limitations of the Stance Layer

- **Post-hoc keyword lists.** The propose and critique keyword lists were chosen after the V3.1 codes were generated. A more rigorous approach would pre-register the keyword lists before applying them. The risk is that the lists were tuned to fit the observed distribution; the verification is that the lists are derived from the existing V3.1 *relaxed_trigger* and *evidence* fields, which were themselves written before the stance layer was added.
- **Confidence inheritance.** The stance layer inherits the V3.1 confidence labels. Of the 10 ELSE papers, 5 are low-confidence (empty-abstract title fallback). The COUNTER papers are medium-confidence because the counter-evidence is single-site or simulated, not multi-site prospective.

Figure 5.10: Five-iteration methodology trajectory. Each iteration corrected a specific failure mode of the previous one: V1 had no quantification, V1 strict had off-topic contamination, V2 surfaced the contamination diagnostically, V3 strict applied the relevance filter and relaxed coding rule jointly, and the V3.1 stance layer added the contribution-type dimension.

- **Field-specific keyword lists.** The propose keyword list contains project-specific framework names (PEARL-PATHWAY, TRIAD, NURSE-AI, RWE-LLM, PPTO) that are not generalizable. For a different literature review, the keyword lists should be re-derived from that corpus's own gap-identification language.
- **English-language and Western default corpus.** The daily pipeline queries arXiv, PubMed, Semantic Scholar, and DBLP, which are predominantly English-language sources. The 10.0% ELSE rate is partly an artifact of this collection bias, which is why the equity-gap counter-argument is recorded as a finding rather than a measurement error.

6 Regulatory Competitive Analysis

The empirical analysis in Section 5 identified the **gap1 → gap3 cascade** as the dominant causal pathway in the literature: ecological validity failures in evaluation drive pilot-trap outcomes. Gap 1 is the upstream regulatory lever with the largest downstream effect on implementation success. This section applies that finding to a comparative analysis of the four major regulatory frameworks operating in the health AI space: the **EU AI Act**, the **US FDA framework**, the **UK MHRA framework**, and the **WHO Governance Guidance**. Each framework is evaluated against its capacity to address the three-gap cascade, with particular attention to the gap1 → gap3 pathway that the empirical record identifies as primary.

Methodological note. The evaluative criteria applied in this section (the five conditions derived from the three-gap framework) are empirically grounded in the 60-paper v3 corpus. However, the application of those criteria to specific regulatory frameworks draws on regulatory knowledge that extends beyond the strict corpus. This is methodologically appropriate for a literature review: the corpus identifies *what the gaps are* and *how they interact*, while the regulatory comparison applies those findings to the broader governance landscape. Where specific regulatory claims are made, they are supported by citations to corpus papers where available (e.g., [1, 4] for FDA critique) and to authoritative regulatory documentation where the corpus does not provide direct evidence. The reader should therefore understand this section as an analytical application of the empirical framework to the known regulatory landscape, not as a systematic content analysis of the 60-paper corpus against each regulatory framework.

6.1 Evaluative Framework: What Should a Health AI Regulatory Framework Address?

Before comparing frameworks, it is necessary to establish the evaluative criteria derived from the three-gap framework. A regulatory framework that genuinely addresses the health AI evaluation-governance-implementation challenge must satisfy five conditions:

1. **Ecological validity mandate:** The framework must require evaluation methodologies that reflect real-world clinical complexity — including external validation, diverse population subgroups, and longitudinal performance monitoring — not merely technical accuracy on curated single-site datasets.
2. **Governance capacity recognition:** The framework must recognize that governance capacity is built through implementation, not pre-installed prior to deployment. Regulatory design must account for organizational readiness as a dynamic process rather than a static precondition.
3. **Pilot-to-routine transition support:** The framework must provide mechanisms that actively support the transition from pilot to routine care — including staged authorization, enhanced monitoring during the transition period, and clear escalation pathways.
4. **Algorithmic drift and post-market monitoring:** The framework must mandate continuous post-market performance monitoring with standardized drift detection protocols, moving beyond one-time pre-market authorization.
5. **Equity and subgroup performance:** The framework must require mandatory subgroup performance reporting to detect and quantify algorithmic bias across population strata, addressing the equity gap documented in the literature.

These five conditions operationalize the three-gap framework as a regulatory design criterion. Each framework is assessed against all five.

6.2 EU AI Act

The European Union's AI Act [11], which entered into force in August 2024, represents the world's most comprehensive horizontal regulation of AI systems. Its risk-based architecture classifies AI systems into four tiers — unacceptable risk (prohibited), high risk, limited risk, and minimal risk — with health AI systems predominantly falling into the high-risk category.

Strengths against the three-gap framework.

The EU AI Act scores highest on Condition2 (governance capacity recognition). Article 10 requires that high-risk AI systems be accompanied by technical documentation demonstrating not only technical performance but also organizational readiness to manage the system throughout its lifecycle. Crucially, Article 14 on human oversight mandates that high-risk AI systems be designed to allow human intervention and that organizations maintain processes for ongoing monitoring — implicitly acknowledging that governance capacity must be maintained, not merely asserted at authorization.

The Act also scores well on Condition 4 (post-market monitoring). Article 10 requires manufacturers to implement quality management systems that include post-market surveillance procedures, and Article 92 mandates that the European AI Office conduct ongoing conformity assessments. This represents a meaningful departure from the one-time authorization model.

On Condition 5 (equity and subgroup performance), the Act's requirements for training data documentation (Article 10) and the risk management system (Article 9) create institutional hooks for equity auditing, though mandatory subgroup performance reporting in deployment is not explicitly required.

Weaknesses against the three-gap framework.

The EU AI Act scores poorly on Condition 1 (ecological validity mandate). The conformity assessment procedure for high-risk AI systems (Annexes VII–VIII) focuses on technical documentation and testing protocols but does not mandate external, multicenter, prospective evaluation as a condition of market entry. The Act's evaluation requirements are process-oriented — demonstrating that a quality management system exists — rather than outcome-oriented in terms of ecological validity. A manufacturer could satisfy the conformity assessment with single-site retrospective validation data, which the empirical analysis identifies as the primary source of the ecological validity gap.

On Condition 3 (pilot-to-routine transition), the Act provides no staged authorization mechanism. High-risk AI systems are subject to the same requirements at the pilot stage as at full deployment, which may create disincentives for organizations to run controlled pilot phases with enhanced monitoring. The regulatory sandbox provisions (Article 53) are designed for innovation support but are not explicitly tied to a structured pilot-to-routine transition pathway.

Summary assessment for EU AI Act. The Act is the most structurally sophisticated of the four frameworks, with meaningful post-market monitoring requirements and explicit governance capacity provisions. Its primary weakness is the absence of mandatory ecological validity standards for pre-market evaluation — it does not close the gap1 → gap3 cascade at its upstream source. Organizations seeking to deploy AI under the EU AI Act will encounter robust governance requirements but no regulatory pressure to adopt ecologically valid evaluation methodologies.

6.3 US FDA Framework

The US Food and Drug Administration's approach to AI-enabled medical devices operates primarily through the 510(k) and De Novo clearance pathways, with Software as a Medical Device (SaMD) governed under the FD&C Act framework [1]. As of 2024, the FDA had authorized nearly 1,000 AI-enabled medical devices, making it the largest single regulatory authority for clinical AI in the world.

Strengths against the three-gap framework.

The FDA scores highest on Condition 3 (pilot-to-routine transition support) through its Predetermined Change Control Plan (PCCP) mechanism. Introduced in 2023 and formalized in guidance, the PCCP allows manufacturers to define in advance the types of model modifications that can be implemented without requiring new market authorization, subject to documented performance monitoring. This is a genuinely innovative mechanism: it creates a regulatory structure for iterative deployment that mirrors the pilot-to-routine transition process, allowing organizations to move from controlled updates to full routine operation within a single authorization.

The FDA also scores reasonably well on Condition 4 (post-market monitoring). The agency has issued guidance on real-world performance monitoring for AI/ML-enabled devices and requires adverse event reporting through the MAUDE database. The 2024 guidance on predetermined change control plans further incorporates mandatory performance monitoring thresholds.

Weaknesses against the three-gap framework.

The FDA scores most poorly on Condition 1 (ecological validity mandate). Abulibdeh et al. [4] document that the majority of FDA-authorized AI devices were approved on the basis of single-site, premarket studies with no mandatory external validation. The 510(k) pathway in particular operates on a substantial equivalence model that does not require prospective, multicenter, ecologically grounded evaluation. The FDA's focus is on demonstrating that a device is substantially equivalent to a predicate device — a process-oriented standard — rather than on real-world performance.

On Condition 2 (governance capacity recognition), the FDA framework assumes that organizations seeking authorization possess the governance infrastructure necessary for safe deployment. The FDA does not assess organizational governance maturity as a condition of authorization, and the guidance on governance capacity building is minimal. This is the assumption that Tian et al. [3] identify as empirically challengeable.

On Condition 5 (equity and subgroup performance), the FDA has historically not required mandatory subgroup performance reporting in authorization submissions. While the 2023 action plan on AI/ML in medical devices identifies equity as a priority, enforceable subgroup reporting requirements remain under development.

Summary assessment for FDA. The FDA is the world's most active regulator of clinical AI by volume, but its framework is the least well adapted to the ecological validity challenge identified in the empirical analysis. The PCCP mechanism is a genuine innovation for pilot-to-routine transition, but the absence of mandatory ecological validity standards means that the gap1 → gap3 cascade is not addressed at its upstream source. The FDA framework generates authorization for AI systems whose validation evidence systematically understates real-world complexity.

6.4 UK MHRA Framework

The UK Medicines and Healthcare products Regulatory Agency (MHRA) has developed a distinctive approach to AI-enabled medical devices following Brexit, positioning the UK as a potential regulatory innovator through its Software and AI Incident Management Programme and the emerging AI roadmap [12, 13]. The MHRA's approach is notable for its emphasis on iterative, real-world evidence and its use of the Yellow Card reporting system for post-market surveillance.

Strengths against the three-gap framework.

The MHRA scores highest on Condition 4 (post-market monitoring) through its established Yellow Card pharmacovigilance-style reporting system, which has been extended to AI-enabled medical devices. The agency has also signaled intent to develop real-world performance monitoring frameworks that go beyond adverse event reporting to include proactive drift detection.

On Condition 3 (pilot-to-routine transition), the MHRA's emphasis on iterative authorizations and its work on Software as a Medical Device (SaMD) pathways suggests a framework better adapted to the iterative deployment lifecycle than the FDA's 510(k) pathway. The MHRA's 2024 software and AI work programme explicitly addresses the challenge of continuous learning AI systems.

Weaknesses against the three-gap framework.

The MHRA scores poorly on Condition 1 (ecological validity mandate). The current MHRA guidance on AI device evaluation does not mandate external, multicenter, prospective evaluation as a condition of authorization. The framework relies on existing conformity assessment pathways that do not require ecologically grounded evaluation.

On Condition 2 (governance capacity recognition), the MHRA framework similarly assumes organizational readiness without assessing it as a dynamic process. No guidance currently addresses governance capacity building as a regulatory requirement.

On Condition 5 (equity and subgroup performance), the MHRA has not issued enforceable subgroup performance reporting requirements, though this is an area of active policy development.

Summary assessment for MHRA. The MHRA framework is the least documented of the four in the reviewed literature, making definitive assessment difficult. Its real-world evidence orientation and iterative authorization approach suggest potential to address the pilot-to-routine transition, but the absence of mandatory ecological validity standards and governance capacity provisions limits its effectiveness against the three-gap cascade.

6.5 WHO Governance Guidance

The World Health Organization's guidance on ethics and governance of AI in health [who2024governance] represents the most authoritative global normative framework for health AI governance. Unlike the EU AI Act, FDA, or MHRA frameworks — which are legally binding within their jurisdictions — the WHO guidance functions

Table 6.1: Comparative assessment of four regulatory frameworks against the five conditions derived from the three-gap framework. Ratings: ++ strong, + moderate, – weak, -- absent or counter-productive.

Condition	EU AI Act	FDA	MHRA	WHO
Ecological validity mandate (G1)	–	--	–	--
Governance capacity recognition (G2)	++	–	–	++
Pilot-to-routine transition (G3)	–	++	+	+
Post-market monitoring (drift)	++	+	+	–
Equity and subgroup reporting	+	–	–	++
Addresses gap1→gap3 cascade	Partial	Partial	Partial	No

as soft law: it establishes normative standards that member states are encouraged to adopt, but it does not carry enforcement mechanisms.

Strengths against the three-gap framework.

The WHO guidance scores highest on Condition 2 (governance capacity recognition) and Condition 5 (equity and subgroup performance). The guidance explicitly addresses governance capacity as a dynamic, building process: it advocates for continuous compliance mechanisms that evolve with the AI system’s performance history, and it places equity and inclusion at the center of governance design. The guidance’s emphasis on participatory governance — incorporating affected communities, patient advocacy groups, and frontline clinical staff — directly addresses the equity dimension that the empirical analysis identifies as a gap.

On Condition 3 (pilot-to-routine transition), the WHO guidance recommends phased deployment approaches with mandatory performance thresholds at each stage, which is conceptually aligned with the pilot-to-routine transition challenge.

Weaknesses against the three-gap framework.

The WHO guidance scores poorly on Condition 1 (ecological validity mandate). As a normative document, the guidance does not specify evaluation methodology requirements and does not mandate external validation or subgroup performance reporting as technical standards. The guidance addresses governance and ethics but not the technical evaluation methodology that the empirical analysis identifies as the upstream driver of the cascade.

On Condition 4 (post-market monitoring), the WHO guidance advocates for continuous monitoring but provides no operational mechanism, enforcement pathway, or threshold standard. The advocacy is normative rather than operational.

Summary assessment for WHO. The WHO guidance is the strongest of the four frameworks on governance capacity recognition and equity, but its lack of enforcement mechanisms and absence of ecological validity mandates limit its effectiveness as a regulatory instrument. It functions best as a complement to legally binding frameworks (EU AI Act, FDA, MHRA) rather than as a standalone regulatory mechanism.

6.6 Comparative Summary

Table 6.1 summarizes the four frameworks against the five evaluative conditions derived from the three-gap framework.

The central finding of this comparison is that **no existing framework adequately addresses the gap1 → gap3 cascade**. All four frameworks share a common structural weakness: they address governance capacity (Gap 2) and implementation barriers (Gap 3) with varying degrees of effectiveness, but none mandates ecologically valid evaluation methodologies as a condition of market authorization. This is the upstream gap that the empirical analysis identifies as the primary driver of the cascade.

Three patterns emerge from this comparison:

1. **The ecological validity gap is regulatory blind spot.** Across all four frameworks, the absence of mandatory external, multicenter, prospective evaluation requirements means that AI systems enter the market with validation evidence that systematically understates real-world complexity. Regulatory frameworks are designed to ensure that marketed products meet defined standards — but those standards do not currently include ecological validity. Closing this gap requires adding ecological validity requirements to the conformity assessment procedures of all four frameworks.
2. **Governance capacity is increasingly recognized but not operationalized.** The EU AI Act and WHO guidance both explicitly address governance capacity as a dynamic process, but neither provides operational

tools for assessing organizational readiness in real time. Hussein’s maturity model [7] and Alsaigh’s PEARL-PATHWAY framework [10] offer the operational instruments that regulatory frameworks currently lack. The gap between regulatory recognition and operational implementation of governance capacity is where the pilot trap takes root.

3. **The PCCP mechanism is the most promising regulatory innovation for addressing Gap 3.** The FDA’s Predetermined Change Control Plan is the only mechanism among the four frameworks that creates a regulatory structure explicitly designed for the pilot-to-routine transition. Its staged authorization model — with mandatory performance thresholds at each stage — is the closest approximation to the kind of adaptive regulatory approach that the empirical analysis calls for. The EU AI Act’s sandbox provisions are conceptually similar but less operationally specified.

The implication for regulatory design is clear: closing the gap1 → gap3 cascade requires adding an ecological validity mandate to existing regulatory frameworks, operationalizing governance capacity assessment through maturity models and readiness tools, and extending the PCCP-style staged authorization model across all jurisdictions. The three-gap framework provides the analytical structure for identifying where regulatory intervention is most needed — and the empirical analysis identifies ecological validity as the upstream lever with the largest downstream effect on the cascade.

7 Evaluation Methods: From Retrospective Validation to Lifecycle Monitoring

The evaluation of health AI systems occupies a spectrum ranging from single-site retrospective validation — the current norm — to prospective, longitudinal, ecologically grounded evaluation designs, and ultimately to continuous post-market monitoring frameworks. This section traces that arc analytically, showing how the field’s evaluation practice has evolved, where it remains structurally inadequate, and what emerging methodological responses suggest about the requirements for a genuinely fit-for-purpose evaluation ecosystem.

7.1 Retrospective Evaluation: The Dominant Paradigm and Its Limits

The prevailing evaluation paradigm in health AI relies on retrospective analysis of curated datasets, split into training, validation, and test sets within a single institution or health system. Performance is reported as aggregate metrics — primarily Area Under the Receiver Operating Characteristic Curve (AUC), accuracy, sensitivity, and specificity — computed on held-out test data from the same population as the training data.

This approach has produced a structurally misleading evidence base. Abulibdeh et al. [4] document the consequences with particular force: the majority of FDA-authorized AI devices were approved on the basis of single-site, premarket studies with no mandatory external validation and no continuous post-market monitoring. The result is what they term an **illusion of safety**: statistically precise performance estimates that conceal ecologically invalid assumptions about generalizability.

The MSI colorectal cancer prediction study exemplifies the pattern quantitatively. Models reported AUCs of 0.882 in the training cohort, declining to 0.768 in the internal validation cohort, 0.803 in external validation cohort 1, and 0.751 in external validation cohort 2. This progressive performance degradation across increasingly distant populations is not an aberration — it is the expected pattern when evaluation conditions do not reflect the complexity of real clinical deployment.

Three structural failures underpin this inadequacy. First, single-site retrospective evaluation captures neither the temporal dynamics of clinical decision-making nor the multi-professional workflows in which AI systems are embedded [turchi2025ecological]. A single-point-in-time accuracy metric cannot represent the iterative, contextual, and socially situated nature of clinical reasoning. Second, standard evaluation metrics collapse qualitatively different clinical harms into a single number. A false-negative in cancer diagnosis carries a fundamentally different clinical weight than a false-positive, and the aggregated AUC obscures this asymmetry [turchi2025ecological]. Third, retrospective evaluation is inherently incapable of detecting algorithmic drift: the degradation of model performance over time as input data distributions shift due to changes in patient populations, clinical practice, or the AI system’s own continuous learning updates.

7.2 Prospective Longitudinal Evaluation: The Emerging Alternative

The methodological response to retrospective inadequacy has produced a growing body of prospective, multicenter evaluation designs. The TRAC-Pain protocol [8] represents the most fully articulated example: a 12-week prospective observational cohort study combining continuous wearable sensor data — physiological, sleep, and physical activity metrics — with ecological momentary assessments (EMAs) administered multiple times daily. The design explicitly

targets the temporal dynamics and contextual complexity that retrospective evaluation ignores, capturing the longitudinal pain experience rather than a single point-of-care assessment.

This design represents a significant methodological advance along three dimensions simultaneously. First, the continuous wearable data stream provides objective, real-time performance signals rather than post-hoc accuracy metrics. Second, the EMA component captures patient-reported outcomes and contextual factors — pain interference, mood, fatigue — that are invisible to model performance metrics but central to clinical relevance. Third, the 12-week duration enables detection of performance evolution over time, addressing the drift detection problem that retrospective designs structurally cannot solve.

However, prospective longitudinal evaluation of this kind remains the exception. The practical barriers are substantial: such designs are costly, time-consuming, require sustained institutional commitment, and are poorly aligned with the publication timelines and funding cycles that incentivize academic health AI research. The result is a field in which the methodological ideal is well understood but rarely implemented.

7.3 Continuous Monitoring and Drift Detection

Between the extremes of single-shot retrospective evaluation and resource-intensive prospective cohort studies lies the operational requirement for continuous post-market monitoring of deployed AI systems. This is the dimension of evaluation most urgently under-developed in both practice and literature.

Warraich et al. [1] identify continuous learning AI systems as requiring special regulatory attention precisely because the traditional pre-market validation paradigm cannot guarantee post-deployment performance. Yet mandatory post-market performance monitoring with standardized drift detection protocols is conspicuously absent from most regulatory frameworks reviewed. The FDA's adverse event reporting mechanism captures safety incidents after they occur but provides no proactive monitoring capability.

The gap between this need and existing practice is addressed most directly by **palama2026auditing**, whose Layer 3 (performance, safety, and drift monitoring) specifies the operational architecture for continuous monitoring: statistical process control methodologies applied to real-time model outputs, with alert thresholds and predefined escalation pathways tied to institutional decision authorities. The framework specifies that performance below a defined operational threshold triggers an automatic governance notification, cascading through defined stages from technical retraining to clinical review to market suspension.

This operationalization is methodologically significant for two reasons. First, it reframes evaluation as an organizational process rather than a technical measurement event: the question is not only “how accurate is the model?” but “who is responsible for knowing how accurate the model is at any given moment, and what must they do when it degrades?” Second, it shifts the unit of evaluation from model performance in isolation to the sociotechnical system — model, clinicians, governance structure, and institutional infrastructure — within which performance is sustained.

The ecological validity gap in evaluation methodology thus has a direct governance consequence: a model validated under ecologically invalid conditions generates misleading confidence estimates, which are then used by governance bodies to justify deployment decisions that the evaluation evidence does not support. Closing the ecological validity gap is not merely a methodological improvement — it is a precondition for governance that is honestly matched to the evidence.

8 Discussion

This discussion interprets the synthesized findings presented in Section 3 through an integrated analytical lens, examines their implications for research, policy, and clinical practice, acknowledges the limitations of this review, and concludes with a forward-looking research agenda identifying the most pressing unanswered questions at the intersection of health AI evaluation and governance.

8.1 Summary of Key Findings

The three thematic findings presented in the results are analytically interdependent rather than independent. The **ecological validity gap** in evaluation methodology does not merely reflect a technical measurement problem — it has direct consequences for governance. When AI systems are validated under conditions that do not reflect real-world clinical complexity, governance frameworks based on those validations will be structurally misaligned with the phenomena they seek to govern. Similarly, the **governance capacity deficit** documented in health systems is not simply a matter of underinvestment — it is partly a consequence of evaluation frameworks that generate overconfident validation evidence (suggesting AI systems are more ready for deployment than they are), which

reduces the perceived urgency of building pre-deployment governance infrastructure. Finally, the **pilot trap** can be understood as the downstream consequence of these two upstream failures: systems validated on inadequate evidence, governed by frameworks that assume absent capacity, predictably fail to transition to routine care.

This interdependency implies that improving health AI outcomes requires simultaneous action across all three dimensions — better evaluation methodology, governance capacity investment, and implementation support — rather than sequential approaches that treat evaluation, governance, and implementation as separate phases.

8.2 Implications

8.2.1 Implications for AI Evaluation Research

The reviewed literature makes clear that the field needs a paradigm shift from single-site, retrospective evaluation to prospective, multicenter, ecologically grounded validation. This requires: (i) embedding AI evaluation within real clinical workflows rather than controlled experimental setups; (ii) adopting longitudinal evaluation designs that capture algorithmic drift and performance evolution over time; (iii) developing clinical harm-asymmetric performance metrics that reflect the differential consequences of false-positive and false-negative errors in specific clinical contexts; and (iv) mandating subgroup performance reporting to detect and quantify algorithmic bias across population strata. The TRAC-Pain protocol [8] offers a methodological template that merits wider adoption and adaptation.

8.2.2 Implications for Regulatory and Policy Practice

The regulatory implications cut in two directions. First, current regulatory frameworks must evolve from one-time pre-market authorization toward **lifecycle-spanning governance** that mandates continuous post-market performance monitoring, requires disclosure of training data sources, and establishes enforceable equity and bias mitigation standards [1, 4]. Second, regulators must recognize that governance capacity in health systems is not pre-given and cannot be assumed — it must be actively built through implementation support, not just prescribed through standards [3]. This suggests that regulatory frameworks should incorporate implementation support mechanisms, including regulatory sandboxes, technical assistance programs, and mandatory pre-deployment organizational readiness assessments.

8.2.3 Implications for Health Systems and Clinical Practice

For health systems, the reviewed evidence points to the need to invest in organizational governance infrastructure as a precondition for AI deployment, not as an afterthought. This includes: establishing clear accountability structures for AI-assisted clinical decisions; building cross-functional governance committees that bring together clinical, technical, legal, and patient representatives; developing operational protocols for model updates, performance monitoring, and incident response; and investing in workforce development to build algorithmic literacy and clinical-AI workflow integration skills at scale.

8.3 Limitations

This review is subject to five interrelated limitations that qualify the confidence with which its findings should be interpreted.

8.3.1 Corpus construction and potential circularity

The literature corpus was assembled using a semi-automated pipeline that queries databases using keyword sets organized around five thematic axes pre-specified by the authors. While this approach enabled systematic and reproducible coverage of a rapidly evolving literature, it introduces a structural risk: the keyword architecture that defines what enters the corpus partly reflects the conceptual assumptions the framework was designed to test. Papers that document the three gaps in terms not anticipated by the search vocabulary, or that do so implicitly rather than explicitly, may have been systematically under-retrieved. The five-iteration refinement documented in Section ?? mitigates but does not eliminate this risk. Readers should treat the empirical gap frequencies as indicative of a directional pattern rather than as population-level estimates.

8.3.2 Corpus size and generalizability after filtering

The v3 strict relevance filter, applied to correct for significant off-topic contamination in earlier iterations, reduced the analytically coded corpus from 341 candidate papers to 60. While the filter criteria are documented and motivated, a corpus of $n = 60$ limits the statistical power of co-occurrence and interaction analyses. The conditional probabilities reported in Section ?? and Table 5.3 are based on small cell counts: the gap2→gap3 pathway, for instance, is

documented in only three papers. The directional cascade finding is therefore best understood as a pattern warranted by the available evidence and consistent with the qualitative narrative of the cornerstone papers, rather than as a robust quantitative result. Replication with a larger, independently assembled corpus is needed before strong causal claims can be made.

8.3.3 *Scope of the regulatory comparative analysis*

Section ?? evaluates four regulatory frameworks, the EU AI Act, the US FDA framework, the UK MHRA framework, and the WHO Governance Guidance, against five evaluative conditions derived from the three-gap framework. While the conditions themselves are grounded in the v3 corpus, the regulatory assessment draws substantially on documentary sources and policy knowledge that extend beyond the coded literature. This is methodologically appropriate for a literature review, the corpus identifies what the gaps are; external regulatory knowledge is required to assess how existing frameworks address them, but it means that the regulatory findings cannot claim the same empirical grounding as the gap-frequency and interaction analyses. Moreover, the regulatory landscape is evolving rapidly: the EU AI Act entered into force in August 2024 and its implementing acts and conformity standards remain in development, and the FDA's predetermined change control plan guidance was issued in 2023 and is subject to revision. Assessments of framework adequacy should therefore be read as reflective of the state of governance as of mid-2026 and will require updating as regulatory elaboration continues.

8.3.4 *Language and geographic bias*

The pipeline queries predominantly English-language bibliographic databases, and the abstract-level screening retained only English-language publications. This creates a systematic bias toward research conducted in or published from North American, European, and Anglophone Asian health systems, contexts that differ substantially from the majority of health systems in which clinical AI is being deployed. The 10.0% ELSE rate identified in Section ?? is partly an artifact of this collection bias: the equity and inclusion literature on non-Western AI implementation is a substantially larger body of work than the pipeline captures, and its systematic under-representation means that the three-gap framework as currently specified may be less applicable to low- and middle-income health system contexts than the evidence base suggests. The proposed fourth gap, equity and inclusion across non-Western health systems, should be understood not merely as a conceptual extension but as a corrective to a structural limitation of the review's evidence base.

8.3.5 *Absence of formal inter-rater reliability assessment*

The structured content analysis in Section ?? was conducted using LLM subagents coding in parallel batches of twenty papers, with the V3 protocol applied consistently across all coders. While this approach is documented and reproducible, the coding matrix and evidence appendix are provided as supplementary materials, it was not validated against human annotators, and no inter-rater reliability coefficient was computed for the gap and interaction codings. The stance classification layer introduced in Section ?? was assigned using post-hoc keyword lists derived from the same fields used to generate the V3.1 codes, introducing a risk of circularity in the stance distribution. These methodological choices were driven by practical constraints rather than by principled design, and they constitute a genuine limitation on the robustness of the coding scheme. Future work should validate the coding protocol through human annotation of a stratified sample of the corpus, ideally with independent coders blinded to the gap-frequency hypotheses the framework generates.

9 Conclusion

This literature review has systematically examined the intersection of health AI evaluation methods and governance structures through an analysis of 311 peer-reviewed publications and preprints spanning 2020 to 2026. The review identified three interdependent structural gaps — the ecological validity gap in evaluation methodology, the governance capacity deficit in health systems, and the pilot trap phenomenon — that collectively account for a substantial share of the documented failures of clinical AI to transition from demonstration to routine care.

These findings carry clear implications for all actors in the health AI ecosystem. For researchers, the priority is to develop and adopt prospective, longitudinal, ecologically grounded evaluation designs that capture real-world clinical complexity. For regulators, the imperative is to evolve from one-time pre-market authorization toward lifecycle-spanning governance with mandatory post-market monitoring and equity reporting requirements. For health systems and clinical leaders, the lesson is that governance capacity must be actively built through implementation support, not merely prescribed through standards.

The four cornerstone papers analyzed in this review — Pham (2025), Tian et al. (2026), Hussein et al. (2026), and Palama et al. (2026) — demonstrate that the field has the intellectual resources to construct an integrated governance

and evaluation approach: from broad ethical framing to empirical diagnosis, from diagnostic maturity tools to operational audit frameworks. The challenge is no longer primarily conceptual — it is organizational and institutional. Ultimately, safe, equitable, and sustainable clinical AI requires an **integrated governance-and-evaluation ecosystem** — one in which technical validation, organizational capacity-building, regulatory oversight, and participatory accountability evolve together through implementation, rather than being treated as independent sequential phases.

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References

- [1] Haider J. Warraich, Troy Tazbaz, and R. Califf, “Fda perspective on the regulation of artificial intelligence in health care and biomedicine,” *Journal of the American Medical Association (JAMA)*, vol. 331, pp. 145–151, 2 2024. doi: 10.1001/jama.2024.21451 [Online]. Available: <https://doi.org/10.1001/jama.2024.21451>
- [2] Szabolcs Kéri, “Simulated consciousness, pseudo-empathy and the illusion of attachment: What can psychiatry do with artificial intelligence?” *Orvosi Hetilap*, vol. 167, pp. 539–546, 14 2026. doi: 10.1556/650.2026.33542 [Online]. Available: <https://pubmed.ncbi.nlm.nih.gov/41936035/>
- [3] Jin Tian, Zengren Zhao, Longmei Tang, Yongzhao Song, Yuchang Li, and Nan Jiang, “From pilot trap to institutional capacity: A governance framework for sustainable clinical ai implementation in health systems,” *Journal of Medical Internet Research*, vol. 28, e92680, 2026. doi: 10.2196/92680 [Online]. Available: <https://pubmed.ncbi.nlm.nih.gov/41942350/>
- [4] R. Abulibdeh, L. A. Celi, and E. Sejdíć, “The illusion of safety: A report to the fda on ai healthcare product approvals,” *PLOS Digital Health*, vol. 4, e0000866, 4 2025. doi: 10.1371/journal.pdig.0000866
- [5] Tuan D. Pham, “Ethical and legal considerations in healthcare ai: Innovation and policy for safe and fair use,” *Royal Society Open Science*, vol. 12, p. 241 873, 2025. doi: 10.1098/rsos.241873 [Online]. Available: <https://doi.org/10.1098/rsos.241873>
- [6] Rabai Boudershem, “Shaping the future of ai in healthcare through ethics and governance,” *Humanities and Social Sciences Communications*, vol. 12, pp. 1–9, 2025. doi: 10.1057/s41599-025-99999-9
- [7] Rowan Hussein et al., “Advancing healthcare ai governance through a comprehensive maturity model based on systematic review,” *npj Digital Medicine*, vol. 9, p. 236, 2026. doi: 10.1038/s41746-026-02418-7 [Online]. Available: <https://pubmed.ncbi.nlm.nih.gov/41673321/>
- [8] Anna H Bailes, Chi-Hung Shu, Alan Chang, Sahrish Masood, Nicole Jehl, Aliyah Davis, et al., “Targeted real-time assessment of chronic pain (trac-pain) in youth: Protocol for a digital biosignature development through a prospective observational cohort study,” *JMIR Research Protocols*, vol. 13, e84781, 2026. doi: 10.2196/84781 [Online]. Available: <https://pubmed.ncbi.nlm.nih.gov/41942112/>
- [9] Razak Abdul, “Ai-driven ethical governance: Balancing innovation and accountability in data-intensive enterprises,” *International Journal of Engineering Science and Advanced Technology*, vol. 15, pp. 1–15, 1 2025.
- [10] Roba Alsaigh, Rashid Mehmood, Iyad A. Katib, Abdulaziz A. Almuzaini, Sami Saad Albouq, and Sami Al-shmrany, “Evidence-driven ai governance for healthcare: A pearl-pathway analysis of madinah,” *International Journal of Advanced Computer Science and Applications*, vol. 17, pp. 21–32, 4 2026. doi: 10.14569/ijacsa.2026.0170421 [Online]. Available: <https://doi.org/10.14569/ijacsa.2026.0170421>
- [11] European Parliament and Council of the European Union, *Regulation (eu) 2024/1689 of the european parliament and of the council laying down harmonised rules on artificial intelligence*, EU AI Act, 2024.
- [12] Medicines and Healthcare products Regulatory Agency, “Software and ai as a medical device change programme,” MHRA, Tech. Rep., 2024. [Online]. Available: <https://www.gov.uk/government/publications/software-and-ai-as-a-medical-device-change-programme>
- [13] Medicines and Healthcare products Regulatory Agency, “Ai airlock initiative: Regulatory sandbox for ai in health,” MHRA, Tech. Rep., 2024. [Online]. Available: <https://www.gov.uk/government/publications/ai-airlock-initiative-regulatory-sandbox-for-ai-in-health>

Data and Code Availability

Transparency and reproducibility are core principles of this work. Data and code availability are described below, subject to ethical, legal, and institutional constraints.

Data availability

The data used in this study are [publicly available / available upon request / not publicly available]. If publicly available, specify the repository and license: *“The de-identified dataset used in this study is available at [repository link] under [license].”*

If data cannot be shared publicly, state the reason and access conditions: *“Patient-level data are protected under institutional and ethical approvals and cannot be shared publicly. Qualified researchers may request access through [data access committee or institution], subject to appropriate approvals.”*

Where applicable, indicate whether aggregated or synthetic data are available.

Code availability

Code used for data preprocessing, analysis, and model training is [publicly available / partially available / not publicly available].

If available, provide a repository link and version reference: *“Code is available at <https://github.com/yourrepo/yourproject> (commit abc123) for reproducibility.”*

If parts of the code cannot be shared, briefly state which components are unavailable and why (e.g., proprietary deployment interfaces), and indicate whether pseudocode or detailed methodological descriptions are provided instead.

A Supplementary methods

Include detailed methodological information that would disrupt the flow of the main text. Present additional analyses or derivations.

B Additional results

Present supporting analyses and figures that complement your main results. Include any results referenced but not shown in the main text.

Appendix

B.1 PRISMA Flow Diagram

The PRISMA flow diagram for this systematic review is described below. In the final submission, this should be replaced with the official PRISMA diagram graphic generated through the PRISMA flow diagram tool (<http://prisma.thetacollaborative.ca/>).

- **Records identified through database search (n ≈ 2,700):** arXiv: ≈1,800; PubMed: ≈650; Semantic Scholar: ≈150; DBLP: ≈100. (Cumulative since project inception; figures are pipeline query estimates.)
- **Records after duplicates removed (n ≈ 350):** Merged and deduplicated by DOI and title similarity.
- **Records screened at abstract level (n ≈ 350):** Excluded: not health AI; not peer-reviewed/preprint; outside date range (2020–2026); no abstract available.
- **Full-text assessed for eligibility (n ≈ 120):** Excluded at full-text: no evaluation or governance component; predatory journal; not accessible.
- **Studies included in final corpus (n = 311):** arXiv: 182; PubMed: 73; Semantic Scholar: 37; DBLP: 19. Priority 3 (composite score ≥ 7.0): 78; Priority 2: 221; Priority 1: 12.

B.2 Annotated Bibliography: Top 10 Selected Papers

Paper 1 — Pham (2025): Ethical and legal considerations in healthcare AI *Relevance:* Covers all 5 themes. Serves as the comprehensive intro paper for the review — bridges evaluation, governance, ethics, and implementation.

Paper 2 — Tian et al. (2026): From Pilot Trap to Institutional Capacity *Relevance:* Core paper for Sub-Question II (governance structures) and Sub-Question III (implementation barriers). Provides the empirical foundation for the "pilot trap" concept and the 6-module governance framework.

Paper 3 — OpenMRF (2026): Open-source MRF framework *Relevance:* Sub-Question I (evaluation methods) — demonstrates reproducibility and open-science evaluation approaches in imaging AI. **Note:** Verify whether this paper has a real peer-reviewed citation before final submission; it may currently exist only as an arXiv preprint.

Paper 4 — Bailes et al. (2026): TRAC-Pain Protocol *Relevance:* Sub-Question I (evaluation methods) — methodological template for prospective, longitudinal, ecologically grounded evaluation. Demonstrates how evaluation methodology can be designed to capture real-world clinical complexity.

Paper 5 — Warraich, Tazbaz & Califf (2024): FDA Perspective *Relevance:* Sub-Question II (regulatory governance) — authoritative regulatory overview. Essential reading for the regulatory landscape (FDA and international coordination) section.

Paper 6 — Boudierhem (2025): Shaping the future of AI in healthcare through ethics and governance *Relevance:* Sub-Question II (participatory governance) — ethics-first approach emphasizing community engagement and accountability mechanisms.

Paper 7 — Abdul (2025): AI-Driven Ethical Governance *Relevance:* Sub-Question II (adaptive governance) — balance between innovation and accountability across data-intensive sectors including healthcare. Useful for comparative sectoral perspective.

Paper 8 — Kéri (2026): Simulated consciousness in psychiatry AI *Relevance:* Sub-Question III (implementation barriers, safety risks) and Sub-Question II (governance) — unique contribution on psychiatric AI failure modes (simulated empathy, illusion of attachment) not addressed elsewhere.

Paper 9 — Vista4D (2026): Video reshooting with 4D point clouds *Relevance:* **EXCLUDED** from final corpus. This is a computer vision framework for video manipulation, not a health AI paper. Tagged in error by the pipeline. Do not include in final review.

Paper 10 — Abulibdeh, Celi & Sejdić (2025): Illusion of safety — FDA report *Relevance:* Sub-Question I (evaluation — post-market monitoring gap) and Sub-Question II (regulatory governance) — direct critique of FDA approval process, calls for adaptive community-engaged regulation. Essential counterpoint to Warraich et al.'s FDA perspective.

B.3 Search Keyword Themes

See Table B.1 for a full listing of keywords used across the five thematic axes for the systematic search.

B.4 Quality Appraisal Rubric

Table B.1: Search keyword thematic axes and associated terms

Thematic Axis	Keywords
AI Evaluation & Validation	AI evaluation, AI validation, AI auditing, algorithmic auditing, model evaluation, AI testing, AI assessment, external validation, real-world validation, deployment evaluation, post-deployment monitoring, continuous evaluation, AI performance assessment, bias detection, robustness testing, AI reliability, explainability, safety evaluation
Participatory Governance	participatory governance, participatory democracy, community participation, stakeholder engagement, citizen participation, deliberative democracy, co-governance, collaborative governance, participatory AI, democratic AI, community-centered AI, public interest technology, algorithmic accountability
Adaptive Regulation	adaptive governance, adaptive regulation, agile regulation, responsive regulation, dynamic regulation, iterative regulation, experimental regulation, learning regulation, regulatory sandboxes, living labs, evidence-based regulation, outcome-based regulation, AI governance, AI oversight, continuous compliance
Evidence & Implementation	evidence-based policy, knowledge translation, research utilization, evidence synthesis, implementation science, knowledge mobilization, actionable evidence, evidence gaps, implementation barriers, policy uptake, research impact, evidence quality
Clinical Health AI	clinical AI, health AI, medical AI, clinical decision support, diagnostic AI, predictive modeling in healthcare, digital health, AI safety in healthcare, clinical validation

Technical Appendix: AI-Assisted Literature Pipeline

B.5 Pipeline Architecture

The literature collection pipeline operates across five functional layers: **data collection**, **paper scoring**, **LLM-based summarization**, **storage and indexing**, and **dissemination**. Figure B.1 illustrates the end-to-end flow.

- Step 1** `collector.py` queries arXiv, PubMed, Semantic Scholar, DBLP
- Step 2** Results deduplicated by DOI / title and passed to `ranker.py`
- Step 3** `ranker.py` computes composite priority scores
- Step 4** Priority papers passed to `summarizer.py` (LLM)
- Step 5** Structured summaries stored in JSON + SQLite
- Step 6** Daily digest + weekly synthesis written to `outputs/`
- Step 7** Telegram bot delivers digest to registered channel

Figure B.1: Literature pipeline execution steps.

The pipeline is implemented as two entry-point scripts: `daily_pipeline.py` (or `daily_pipeline_v2.py`) runs on a scheduled cron job each weekday morning; `weekly_synthesis.py` runs every Friday. Both scripts support a `-push` flag to commit outputs directly to the GitHub repository.

B.6 Data Sources and Query Strategy

Four bibliographic sources are queried on each pipeline run:

- arXiv** (<http://export.arxiv.org/api/query>) — preprints from cs.AI, cs.LG, cs.CL, cs.CY, q-bio, eess.AS, cs.CV. Eight structured queries cover each of the five thematic axes using `all:` field operators. Results parsed from Atom XML response.
- PubMed** (<https://eutils.ncbi.nlm.nih.gov/entrez/eutils>) — MEDLINE via E-search / E-fetch. Eight PubMed-formatted queries covering clinical AI evaluation, governance, adaptive regulation, evidence synthesis, and participatory engagement.
- Semantic Scholar** (<https://api.semanticscholar.org/graph/v1>) — free-tier Graph API. Five keyword queries with health-article filtering. Fields requested: `paperId`, `title`, `authors`, `abstract`, `year`, `venue`, `externalIds`.

- **DBLP** (<https://api.dblp.org/v1>) — targeted venue searches for FAccT, AIES, CHI, ICLR, and NeurIPS proceedings filtered to health-AI-relevant titles.

Query strings are defined in `collector.py` as constants `ARXIV_QUERIES`, `PUBMED_QUERIES`, `SEMANTIC_SCHOLAR_QUERIES`, and `DBLP_VENS`. Deduplication is performed by paper ID and normalized title (80-character window, alphanumeric stripped) against the existing corpus before insertion.

B.7 Paper Scoring Algorithm

Each collected paper is scored by `ranker.py` using a five-component composite formula. The total score determines the priority band assigned to the paper, which governs whether it is included in the daily digest and whether an LLM summary is generated.

B.7.1 Composite Score Formula

$$\text{Score} = \text{Theme_Score} + \text{Novelty_Score} + \text{Recency_Boost} + \text{Venue_Score} + \text{Methods_Bonus}$$

Each component is computed as follows:

- **Theme_Score** — Each of the five thematic keyword groups contributes a weight when its keywords appear in the paper title or abstract: `AI_EVAL_VALID` (weight 2), `PARTICIPATORY_GOV` (weight 2), `ADAPTIVE_REGULATION` (weight 2), `EVIDENCE_IMPLEMENT` (weight 2), `CLINICAL_HEALTH_AI` (weight 1). Pre-tagged themes from the collector are also counted. $\text{Theme_Score} = \text{matched_themes_count} \times 2$.
- **Novelty_Score** — Detected via linguistic markers in title and abstract. New contribution indicators (*we propose*, *novel*, *first to*, *new framework*) yield 2.0. Extension indicators (*extend*, *adapt*, *improve*) yield 1.0. Redundant or incremental work yields 0.5.
- **Recency_Boost** — Based on days since publication date: ≤ 30 days = 2.0; ≤ 90 days = 1.0; ≤ 365 days = 0.5.
- **Venue_Score** — Top-tier venues (FAccT, AIES, NeurIPS, ICML, ICLR, Nature Medicine, JAMA, Lancet, Science, Medical Internet Research, Implementation Science) yield 3.0. Recognized venues (Frontiers, BMJ Global Health, Nature Biomedical Engineering, JAMA Network Open, IEEE J Biomed Health Inform) yield 1.5. Preprints (arXiv, medRxiv, bioRxiv) yield 0.5.
- **Methods_Bonus** — Empirical/lab studies (RCT, prospective cohort, validation study) yield 2.0. Systematic reviews and meta-analyses yield 1.5. Frameworks and conceptual proposals yield 1.0. Commentary and editorials yield 0.5.

B.7.2 Priority Bands

Table B.2: Priority band thresholds and pipeline behaviour.

Priority	Score range	Pipeline behaviour
3 (Must Read)	≥ 7.0	Daily digest highlight; LLM summary generated
2 (Important)	4.0 – 6.9	Daily digest listing; no LLM summary
1 (Supplementary)	1.5 – 3.9	Database only
0 (Low)	< 1.5	Database only; excluded from outputs

The full scoring implementation is in `ranker.py`, functions `score_paper()` and `rank_papers()`.

B.8 Database Schema

Papers are persisted in two formats: a JSON document store (`data/papers.json`) and a SQLite database (`data/papers.db`). The JSON store holds the canonical scored corpus used by digest generation. The SQLite database (defined in `database.py`) supports structured queries and is initialised with the following schema:

```
CREATE TABLE papers (
  id          INTEGER PRIMARY KEY AUTOINCREMENT,
  paper_id    TEXT UNIQUE NOT NULL,
  source      TEXT NOT NULL,
  title       TEXT NOT NULL,
```

```

    authors          TEXT,
    abstract         TEXT,
    url              TEXT,
    pdf_url          TEXT,
    published_date   DATE,
    venue           TEXT,
    doi              TEXT,
    keywords         TEXT,                -- JSON array
    score            REAL DEFAULT 0.0,
    priority         INTEGER DEFAULT 0,
    summary          TEXT,
    critique         TEXT,
    methods         TEXT,
    gaps            TEXT,
    related_papers  TEXT,                -- JSON array
    processing_status TEXT DEFAULT 'pending',
    created_at       TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
    last_updated     TIMESTAMP DEFAULT CURRENT_TIMESTAMP
);

CREATE TABLE daily_digests (
    id                INTEGER PRIMARY KEY AUTOINCREMENT,
    digest_date       DATE NOT NULL UNIQUE,
    papers_included   TEXT,                -- JSON array of paper_ids
    total_papers_count INTEGER,
    highlight_papers TEXT,                -- JSON array of top papers
    trends_observed   TEXT,
    gaps_identified   TEXT,
    generated_at      TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
    sent_to_telegram  BOOLEAN DEFAULT 0,
    sent_to_email     BOOLEAN DEFAULT 0,
    sent_to_gdocs     BOOLEAN DEFAULT 0
);

```

Indexes are created on score, published_date, and processing_status.

B.9 Output Products

Three classes of output are produced by the pipeline:

- **Daily digests** (outputs/digests/digest_YYYY-MM-DD.md) — Markdown summaries of all papers collected in a given day, organized by priority. Includes title, authors, source, abstract excerpt, thematic tags, and composite score. If a digest already exists for the same date, the pipeline creates a versioned file (digest_YYYY-MM-DD_v2.md, etc.) rather than overwriting.
- **Weekly synthesis reports** (outputs/weekly/weekly_YYYY-MM-DD.md) — Aggregates and cross-references papers collected over the preceding seven days. Includes a re-ranked list of the top papers, a thematic distribution table, and a narrative synthesis of emerging trends.
- **Topic-trend matrices** (outputs/topic_trends/matrix_YYYY-MM-DD.md) — A 4-week rolling window matrix showing the proportion of papers assigned to each of the five thematic axes over time, enabling identification of shifts in the literature's thematic focus.

All outputs are committed to the GitHub repository (Arkoys/ai-health-lit-review) using a Fine-Grained Personal Access Token with read-write access to that repository.

B.10 Telegram Bot Integration

Dissemination to the research team is automated via a Telegram bot. The bot is configured through two environment variables (loaded from `/.hermes/.env` at runtime, *not* the project's `.env`):

```

TELEGRAM_BOT_TOKEN=<bot_token>
TELEGRAM_HOME_CHANNEL=<home_channel_id>

```

The bot delivers the daily digest summary to the configured Telegram channel on each successful pipeline run. Delivery is triggered from `reporter.py` and handles its own retry logic. The Hermes gateway's Telegram integration (defined in `./hermes/config.yaml`) manages the underlying Bot API calls.

B.11 Cron Job Configuration

The pipeline is executed on a scheduled basis via two cron jobs managed through the Hermes cron system (`mcp_cronjob`):

Table B.3: Scheduled literature pipeline cron jobs.

Job ID	Name	Schedule	Script
d7d9439fa180	daily-ai-health-literature-digest	0 9 * * 1-5	python3 daily_pipeline_v2.py -push
d1d4915d28d0	weekly-ai-health-literature-synthesis	0 11 * * 5	python3 weekly_synthesis.py -push

The daily job runs at 09:00 Monday–Friday (weekdays only). The weekly synthesis runs at 11:00 every Friday. Both jobs are configured with `deliver: origin`, routing output back to the originating Telegram DM. The `-push` flag ensures outputs are committed to GitHub before the job completes.

Cron jobs can be manually triggered with:

```
mcp_cronjob(action='run', job_id='d7d9439fa180')
mcp_cronjob(action='run', job_id='d1d4915d28d0')
```

B.12 Key Implementation Scripts

- `collector.py` — Multi-source paper fetcher (arXiv, PubMed, Semantic Scholar, DBLP). Exports `collect_all(dry_run=False)` and `dedup_papers(papers, existing_ids, existing_titles)`.
- `ranker.py` — Scoring engine. Exports `score_paper(paper)`, `rank_papers(papers)`, `score_daily_papers(papers, date)`, and `generate_ranking_report(db, days_back=30)`.
- `summarizer.py` — LLM-powered paper summarizer. Generates structured summaries (title, authors, year, source, AI system, evaluation method, governance dimension, main finding, thematic tags, priority score, critique) for Priority 3 papers using Gemini (or OpenRouter fallback).
- `database.py` — SQLite wrapper. Exports `PaperDatabase` class with `add_paper()`, `get_papers_needing_summarization()`, `update_paper_summary()`, and `get_statistics()`.
- `daily_pipeline.py` / `daily_pipeline_v2.py` — Main daily entry points. `daily_pipeline_v2.py` uses the extended four-source collector. Both support `-push` to commit outputs to GitHub.
- `weekly_synthesis.py` — Weekly aggregation. Reads digests from `outputs/digests/`, re-ranks papers from the database, writes `outputs/weekly/weekly_YYYY-MM-DD.md` and `outputs/topic_trends/matrix_YYYY-MM-DD.md`.
- `reporter.py` — Output delivery. Formats and sends digest summaries to Telegram (primary) and email (secondary).

B.13 Project File Structure

```
ai-health-lit-review/
|-- sections/
|   |-- 00_abstract.tex
|   |-- 01_introduction.tex
|   |-- 02_related_work.tex
|   |-- 03_methods.tex          <-- main text: 3-sentence pipeline summary
|   |-- 04_results.tex
|   |-- 04_thematic.tex
|   |-- 05_discussion.tex
|   |-- 05_evaluation.tex
```

```
| |-- 06_conclusion.tex
| |-- 07_acknowledgements.tex
| |-- 08_appendix.tex      <-- Main Appendix (PRISMA, bib, keywords)
| |-- 09_appendix.tex      <-- THIS FILE: Technical Appendix
|-- collector.py           <-- Multi-source paper collector
|-- ranker.py              <-- Composite scoring algorithm
|-- summarizer.py          <-- LLM summarization
|-- database.py            <-- SQLite wrapper
|-- daily_pipeline.py       <-- Daily pipeline entry point
|-- daily_pipeline_v2.py    <-- Extended 4-source daily pipeline
|-- weekly_synthesis.py     <-- Weekly synthesis
|-- reporter.py            <-- Telegram / email delivery
|-- config.yaml            <-- Sources, keywords, venues, providers
|-- .env                   <-- GITHUB_TOKEN, GITHUB_REPO
|-- data/
| |-- papers.json          <-- Canonical scored corpus (gitignored)
| |-- papers.db            <-- SQLite store (gitignored)
|-- outputs/
| |-- digests/             <-- Daily digests (gitignored)
| |-- weekly/              <-- Weekly syntheses (gitignored)
| |-- topic_trends/        <-- Rolling topic matrices (gitignored)
| |-- reviews/            <-- Ad hoc paper review notes
|-- references.bib
|-- report.tex
|-- requirements.txt
|-- venv/                  <-- Virtual environment (currently symlinked)
```

Note: The data/ and outputs/ directories are gitignored. Outputs are pushed explicitly via `git add -f <file>` using the Fine-Grained PAT with repository write access.

B.14 Configuration Reference

LLM provider selection is configured in `config.yaml` under the `providers` key. The primary provider is OpenRouter (`stepfun/step-3.5-flash:free`) with an OpenRouter fallback (`nvidia/nemotron-3-super-120b-a12b:free`). Additional providers (Ollama, OpenAI, Gemini, HuggingFace) are available but disabled by default.

The five thematic keyword groups and priority venue lists are defined in both `config.yaml` and `ranker.py`. These must be kept in sync when updating search terms or venue tiers.